Laborbuch SynBio

Content

1 Media and stock solutions2
1 % Agarose solution2
0.9 % NaCl solution2
10 % Glycerol2
Kanamycin stock [50 mg/mL]2
Ampicillin stock [100 mg/mL]2
Trace Element Solution (1000x) (TES)2
Protocatechuic acid stock (PCA)2
CGXII-base (Keilhauer et al.)3
CGXII-Medium with 4 % glucose3
M9-Medium3
LB-Medium4
2xTY Medium4
BHI-Medium4
BHIS-Medium4
2 Methods5
E. coli transformations5
Elektroporation: C. glutamicum transformation protocol7
OneTaq Polymerase protocol (for Colony PCR)10
NEBuilder HiFi DNA Assembly Reaction Protocol13
Protocol for Dephorphorylation of 5‘-ends of DNA using Antarctic Phosphatase13
Rates and Balances13
3 Lab protocol14
3 Primer and plasmids109
Primer109
Plasmids111

1 Media and stock solutions

1 % Agarose solution

Dissolve 1 g Agarose in 100 mL 1xTAE buffer by heating in a microwave until clear solution. Store at 70 °C.

0.9 % NaCl solution

Dissolve 9 g NaCl in 1 L ddH_2O, autoclave.

10 % Glycerol

Glycerol	50 mL

Fill to 500 mL with ddH_2O and autoclave.

Kanamycin stock [50 mg/mL]

Dissolve 0.5 g Kanamycin sulfate in 10 mL ddH_2O, filter-sterilize, make aliquots and store at -20 °C.

Ampicillin stock [100 mg/mL]

Dissolve 0.5 g ampicillin (sodium salt) in 5 mL ddH_2O, filter-sterilize, make aliquots and store at - 20 °C.

Trace Element Solution (1000x) (TES)

$FeSO_4$ *7 H_2O	1 g
$MnSO_4$ *H_2O	1 g
$ZnSO_4$ *7 H_2O	0.1 g
$CuSO_4$ *6 H_2O	0.0313 g
$NiCl_2$ *6 H_2O	0.002 g

Fill to 100 mL with ddH_2O. Set pH 1 with conc. HCl and filter-sterilize. Store at 4 °C.

Protocatechuic acid stock (PCA)

Protocatechuic acid	0.3 g/10 mL

Dissolve in ~1 mL 1 N NaOH, fill to 10 mL with ddH_2O and filter-sterilize. Make aliquots and store at -20 °C.

CGXII-base (Keilhauer et al.)

$(NH_4)_2SO_4$	20 g/L
Urea	5 g/L
KH_2PO_4	1 g/L
K_2HPO_4	1 g/L
MOPS	42 g/L

Dissolve in 700 mL ddH_2O, set pH 7.2 with 5 M KOH, fill up to 920 mL with ddH_2O and autoclave. After autoclaving, add:

1 mL $CaCl_2$ *2 H_2O stock (1000x) [Dissolve 665 mg in 50 mL ddH_2O]
1 mL $MgSO_4$ *7 H_2O stock (1000x) [Dissolve 12.5 mg in 50 mL ddH_2O]
1 mL Biotin stock (1000x) [Dissolve 10 mg in 50 mL ddH_2O]

CGXII-Medium with 4 % glucose

CGXII-base	920 mL
50 % Glucose stock	80 mL
PCA (1000x)	1 mL
TES (1000x)	1 mL

M9-Medium

5x M9 stock:	Na_2HPO_2 *2 H_2O	21.18 g
	KH_2PO_4	7.5 g
	NaCl	1.25 g
	NH_4Cl	2.5 g
	Fill to 500 mL with ddH_2O and autoclave.	
1 M $CaCl_2$ stock:	$CaCl_2$ *2 H_2O	14.7 g
	Fill to 100 mL with ddH_2O and filter-sterilize.	
1 M $MgSO_4$ stock:	$MgSO_4$ *7 H_2O	24.65 g
	Fill to 100 mL with ddH_2O and filter-sterilize.	

For 50 mL M9 Medium, mix:

10 mL	5x M9 stock
40 mL	ddH_2O
5 µL	1 M $CaCl_2$ stock

100 µL	1 M $MgSO_4$ stock
400 µL	50 % Glucose solution

LB-Medium

Yeast Extract	5 g/L
Tryptone	10 g/L
NaCl	10 g/L

Dissolve in ddH_2O, fill to 1 L and autoclave. For agar plates, add 18 g/L Agar-Agar before autoclaving. Pour plates right after.

2xTY Medium

Tryptone	16 g/L
Yeast Extract	10 g/L
NaCl	5 g/L

Dissolve in ~900 mL ddH_2O, set pH 7.0 with 5 M NaOH, adjust to 1 L with ddH_2O and autoclave. For agar plates, add 18 g/L Agar-Agar before autoclaving.

BHI-Medium

Brain-Heart-Infusion	37 g/L

Dissolve in ddH_2O, fill to 1 L and autoclave.

BHIS-Medium

Brain-Heart-Infusion	37 g/L
Sorbitol	91 g/L

Dissolve BHI and Sorbitol in ddH_2O, fill to 800 mL and 200 mL, respectively, and autoclave. Mix after autoclaving.

2 Methods

E. coli transformations

SOP Documentation

Action	Author(s)	Date	Comments Indicate all changes
First draft	Melanie Filbig	27.06.2024	
Review	Marvin Höninger	11.10.2024	

Chemically competent cells (for storage)

Material

- 2TY medium
- flasks for pre- and main-culture
- ice-cold 0.1M $CaCl_2$
- ice-cold 0.1M $CaCl_2$ + 15 % (v/v) glycerol
- 50 ml falcons
- labeled tubes
- liquid N_2

Procedure

- select appropriate *E. coli* strain for application (overview: https://www.thermofisher.com/de/de/home/life-science/cloning/cloning-learning-center/invitrogen-school-of-molecular-biology/molecular-cloning/transformation/competent-cell-genotypes-genetic-markers.html)
- prepare overnight (ON) culture of *E. coli* in 2TY medium
- inoculate main culture (200 ml 2TY in 1L flask) with 500 µl ON culture
- grow at 37°C to OD_{600} 0.5
- spread to 50 ml falcons
- centrifuge 5000 rpm ↻, 10 min ⏲, 4 °C
- discard supernatant
- resuspend pellet in 15 ml ice-cold 0.1M $CaCl_2$
- centrifuge 5000 rpm ↻, 10 min ⏲, 4 °C

Volume can be adjusted according to the number of aliquots required!

- discard supernatant
- resuspend pellet in 15 ml ice-cold 0.1M $CaCl_2$
- centrifuge 5000 rpm ↻, 10 min ⏲, 4 °C
- discard supernatant
- resuspend pellet in 2.5 ml ice-cold 0.1M $CaCl_2$ + 15 % glycerol
- aliquot 200 µl in sterile eppis, flash-freeze in liquid N_2
- store at -70°C

Heatshock transformation

Material

- chemically competent cells
- ice
- heatblock @42°C
- SOC medium (LB also works) (see Sidharth SOP → Link)
- antibiotics-containing agar plates

Procedure

- thaw competent cells on ice
- add DNA
- flick tube to mix (DO NOT VORTEX)
- place on ice for 30 minutes
- heat-shock: 42°C for 60-90 seconds
- place on ice for 2 min
- add 800 µl SOC (LB)
- incubate at 37°C while shaking for 1-2h
- plate 100 µl on agar plate
- centrifuge rest, plate pellet on agar plate
- incubate plates at 37°C overnight

Electrically competent cells (for direct use)

Material

- LB medium (see Sidharth SOP → Link)
- flasks for pre- and main-culture

- ice-cold 10 % (v/v) glycerol

Procedure

- prepare ON culture of *E. coli* in LB
- inoculate main culture to OD_{600} = 0.05
- grow at 37°C to OD_{600} 0.4-0.6
- pellet 4-6 ml culture
- wash cells 3x w/ ice-cold 10 % glycerol
- resuspend in 200 µl ice-cold 10 % glycerol

Electroshock transformation

Material

- electro-competent cells
- electroporation cuvette 2mm
- LB medium (see Sidharth SOP → Link)
- antibiotics-containing agar plates

Procedure

- add 100 ng DNA to 200 µl electro-competent cells
- electroporate: 2mm, 2.5 kV
- add 800 µl LB
- incubate @37°C while shaking for 1h
- plate 100 µl on agar plate
- centrifuge rest, plate pellet on agar plate
- incubate plates at 37°C overnight

Elektroporation: C. glutamicum transformation protocol

SOP Documentation

Action	Author(s)	Date	Comments Indicate all changes

First draft	Marvin Höninger	19.07.2024	

Material:

- **For transformation:**
  - BHI medium
  - BHIS medium
  - BHI plates
  - $BHI_{Kan7.5}$ plates
  - BHI_{Suc} plates
  - EPBI solution
  - EPBII solution
  - Ampicillin solution (5 mg/ml)

- **For the control:**
  - *OneTaq 2x* MasterMix for colony PCR

C. glutamicum transformation protocol

Preculture preparation:

- Pick a colony from a plate and inoculate it in a glass tube containing 5 mL of BHI media. Incubate overnight at 30 °C, shaking conditions.

Main culture:

- Inoculate 150 µL of the preculture in a glass tube containing 4 mL of BHI+sorbitol. Incubate at 30 °C shaking conditions until OD 0.6 is reached.
- Add 4 µL of ampicillin solution (5 mg/mL) and incubate for 1 hour.

Electroporation:

- Transfer 2 mL of cell culture to a new Eppendorf tube and collect the cells from the main culture centrifuging for 15 s at 14.000 rpm 4 °C. Repeat this twice to collect all the cells from the main culture.

- Wash the cells 3 times using the EPBI buffer. To wash the cells, centrifuge them for 15 s 14.000 rpm, remove the supernatant, and resuspend in 1 mL of fresh buffer.
- After the last wash, remove all the EPBI buffer and resuspend the cells in 200 µL EPBII buffer.

NOTE: All the buffers must be ice-cold, and the cells must be kept as cool as possible.

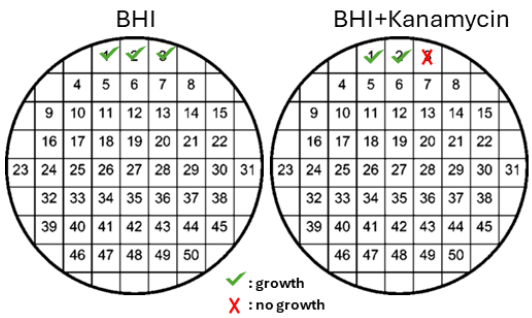
- Transfer the cells to the electroporation cuvette (green cap).
- Add 1 µL of standard miniprep plasmid.
- Start electroporation with the following settings: 200 ohm, 25 µF, 2.5 kV.

Second culture:

- Transfer the cells into an Eppendorf tube containing 1 mL of prewarmed (46 °C) BHI+sorbitol.
- Incubate for 6 minutes at 46 °C on a thermoshaker **STATIC MODE**.
- Transfer the cells to a glass tube containing 1 mL of BHI+sorbitol (final volume ≈ 2.2 mL).
- Incubate overnight at 30 °C, shaking conditions.

Plating:

- Collect all the cells from the cell culture via short centrifugation and plate them on a BHI+Kanamycin plate (7.5 mg/mL).
- Incubate the plates at 30 °C and wait for colony formation (roughly two days).
- Streak single colonies from the BHI+Kanamycin plate to a BHI+sucrose plate.
- Incubate the plates at 30 °C and wait for colony formation (roughly two days).



- Pick single colonies from the BHI+sucrose plate and streak them on both BHI and BHI+Kanamycin plates. These plates must have a grid that allows the identification of the colonies streaked, in this way the **colonies that grow on BHI and don't grow on BHI+Kanamycin** can be picked for colony PCR.

Media, plates, and buffers:

- ***Brain-Heart Infusion (BHI) media***: dissolve 37 g of BHI powder in 1 L of demineralized water, then autoclave. For 2x BHI dissolve 74 g of BHI powder in 1 L of demineralized water.

- ***Brain-Heart Infusion (BHI) agar plates***: dissolve 37 g of BHI powder in 1 L of demineralized water, add 18 g of agar and then autoclave. For antibiotic-containing plates, prepare the antibiotic stock apart and add it before pouring the medium into the plates. Note: most of the antibiotics are heat sensitive, so let the media cool down after autoclavation. For 2x BHI plates add 36 g of agar to the media before autoclavation.

Note: when autoclaving agar-containing solutions, then there is always the risk of something boiling over. If you are unlucky, the agar may then cool down in the lines of the autoclave and solidify, thus blocking the lines. It is therefore advisable to fill a 1L bottle with a maximum of 0.5 L.

- ***1M Sorbitol 1 L solution***: dissolve 182.17 g of sorbitol in 0.5 L demineralized water, bring the volume to 1L and autoclave.

- ***BHI+sorbitol***: mix 1 M sorbitol solution 1:1 with 2x BHI media.

- ***20 % Sucrose 1L solution***: dissolve 200 g of sucrose in 0.5 L demineralized water, bring the volume to 1L, and autoclave.

- ***BHI+sucrose***: mix 20 % Sucrose solution 1:1 with 2x BHI agar, and then pour it into the plates.

- ***EPBI 0.5 L solution***: dissolve in demineralized water 20 mL of HEPES 0.5 M at pH 7.2 and 25 mL of glycerol bring the volume to 0.5 L adding water, then autoclave.

- ***EPBII 0.2 L solution***: dissolve in demineralized water 2 mL of HEPES 0.5 M at pH 7.2 and 30 mL of glycerol bring the volume to 0.2 L adding water, then autoclave.

OneTaq Polymerase protocol (for Colony PCR)

SOP Documentation

Action	Author(s)	Date	Comments Indicate all changes
First draft	Marvin Höninger	19.07.2024	
Review	Sidharth Jayasankar	27.8.2024	

One*Taq* DNA Polymerase is an optimized blend of *Taq* and Deep Vent® DNA polymerases for use with routine and difficult PCR experiments.

- Obtain high yields across a wide range of AT- and GC- content
- 2X higher fidelity than *Taq*
- Master mix format contains inert tracking dye for easy and direct loading of PCR products onto gels

- Product: NEB, catalog # M0486
- https://www.neb.com/en/products/m0486-onetaq-quick-load-2x-master-mix-with-standard-buffer

Quick Protocol for OneTaq® Quick-Load® 2X Master Mix with Standard Buffer

- Assemble the following PCR reaction

We recommend assembling all reaction components on ice and quickly transferring the reactions to a thermocycler preheated to the denaturation temperature (94°C).

Component	25 µl reaction	50 µl reaction	Final Concentration
10 µM Forward Primer	0.5 µl	1 µl	0.2 µM
10 µM Reverse Primer	0.5 µl	1 µl	0.2 µM
Template DNA	variable	variable	< 1,000 ng
One*Taq* Quick-Load 2X Master Mix with Standard Buffer	12.5 µl	25 µl	1X
Nuclease-free water	to 25 µl	to 50 µl	

Notes: Gently mix the reaction. Collect all liquid to the bottom of the tube by a quick spin if necessary. Overlay the sample with mineral oil if using a PCR machine without a heated lid.

Transfer PCR tubes to a PCR machine and begin thermocycling:

Thermocycling conditions for a routine PCR:

STEP	TEMP	TIME
Initial Denaturation	94°C	30 seconds
30 Cycles	94°C 45-68°C 68°C	15-30 seconds 15-60 seconds 1 minute per kb
Final Extension	68°C	5 minutes
Hold	4-10°C	

- General Guidelines

Template:
Use of high quality, purified DNA templates greatly enhances the success of PCR. Recommended amounts of DNA template for a 50 µl reaction are as follows:

DNA	AMOUNT
DNA Genomic	1 ng–1 µg
Plasmid or Viral	1 pg–10 ng

Modification for *Corynebacterium glutamicum*:

Step		Temp	Time
Initial Denaturation		94 °C	**16 min***
30 Cycles	Denaturation	94 °C	15-30 sec
	Annealing	45-68°C	15-60 sec
	Elongation	68 °C	60 sec/kb
Final Extension		68 °C	5 min
Hold		4 °C	∞

*longer initial denaturation

- Primers:

  Oligonucleotide primers are generally 20–40 nucleotides in length and ideally have a GC content of 40–60%. Computer programs such as Primer3 can be used to design or analyze primers. The final concentration of each primer in a PCR may be 0.05–1 µM, typically 0.2 µM.

- Mg^{++} and Additives: Mg^{++} concentration of 1.5–2.0 mM is optimal for most PCR products generated with One*Taq* DNA Polymerase. The final Mg^{++} concentration in 1X One*Taq* Quick-Load Master Mix with Standard Buffer is 1.8 mM. This supports satisfactory amplification of most amplicons. However, Mg^{++} can be further optimized in 0.2 mM increments using $MgCl_2$ (NEB #B9021).

  For amplification of difficult targets, like GC-rich sequences, we recommend One*Taq* Quick Load 2X Master Mix with GC Buffer (NEB# M0487). Alternatively, DMSO or formamide may be used.

- Denaturation:

  An initial denaturation of 30 seconds at 94°C is sufficient to amplify most targets from pure DNA templates. For difficult templates such as GC-rich sequences, a longer denaturation of 2–4 minutes at 94°C is recommended prior to PCR cycling to fully denature the template. Alternatively, use One*Taq* Quick-Load 2X Master Mix with GC Buffer. With colony PCR, an initial 2–5 minute denaturation at 94°C is recommended to lyse cells.

  During thermocycling a 15–30 second denaturation at 94°C is recommended.

- Annealing:

  The annealing step is typically 15–60 seconds. Annealing temperature is based on the T_m of the primer pair and is typically 45–68°C. Annealing temperatures can be optimized by doing a temperature gradient PCR starting 5°C below the calculated T_m. We recommend using NEB's

Tm Calculator to determine appropriate annealing temperature for PCR.

- Extension:

  The recommended extension temperature is 68°C. Extension times are generally 1 minute per kb. A final extension of 5 minutes at 68°C is recommended.

- Cycle Number:

  Generally, 25–35 cycles yields sufficient product. Up to 45 cycles may be required to detect low copy number targets.

- 2-step PCR:

  When primers with annealing temperatures of 68°C or above are used, a 2-step thermocycling protocol (combining annealing and extension into one step) is possible.

- PCR Product:

  The majority of the PCR products generated using One*Taq* DNA Polymerase contain dA overhangs at the 3´ end; therefore the PCR products can be ligated to dT/dU-overhang vectors.

NEBuilder HiFi DNA Assembly Reaction Protocol

See NEBs Protocol: https://www.neb.com/en/protocols/2014/11/26/nebuilder-hifi-dna-assembly-reaction-protocol?pdf=true

2 Fragment Assembly was done by incubating 60 instead of 15 minutes at 50 °C.

Protocol for Dephorphorylation of 5‘-ends of DNA using Antarctic Phosphatase

See NEBs Protocol: https://www.neb.com/en/protocols/0001/01/01/vector-dephosphorylation-protocol?pdf=true

In this case, 5 units of Antarctic Phosphatase corresponds to 1 µL.

Rates and Balances

For comparison of microbial growth and production, the following rates and yields were evaluated:

Name	Symbol	Unit	Determination
Maximal growth rate	μ_{max}	h^{-1}	Slope of CDW-time-curve in exponential phase $\mu_{max}=\frac{1}{c_x}\ \frac{dc_x}{dt}$
Substrate-uptake rate	q_S	$g*g^{-1}*h^{-1}$	$q_S=\frac{1}{c_x}\ \frac{dc_S}{dt}$
Biomass-Substrate-Yield	Y_{XS}	$g_{CDW}*g^{-1}$	$Y_{XS}=\frac{\mu}{q_S}$
Product-Substrate-Yield	Y_{PS}	$g*g^{-1}$	$Y_{PS}=\frac{q_P}{q_S}$

3 Lab protocol

Date	09.10.2024
Who	Noemi, Anna
What	Media produced: • 500 mL LB liquid medium prepared and autoclaved • 500 mL LB agar prepared, after autoclaving 0.5 mL Amp100 added and plates poured • 100 mL 15% Glycerol +0.1 M $CaCl_2$ produced and autoclaved • 100 mL 10% Glycerol prepared and autoclaved • 500 mL 2x TY Medium prepared and autoclaved • Amp100 stocks manufactured, sterile filtered • Kan50 stocks manufactured, sterilfiltered Pre-culture for the production of competent E. coli • 2x 10 mL 2xTY in 100 mL flask • Inoculated with 50 µL E. coli eachα λpir • Incubated at 37 °C

Date	10.10.2024
Who	Alex, Noemi
What	Competent E. coliα λ pir manufactured: • OD the pre-culture: VK1: 10; VK2: 9 • Main culture 1&2 inoculated at 11:35 AM • Inoculated main culture 2 additionally with 1.5 mL of the pre-culture at OD 0.1 at 12 o'clock • OD measurement of the main crops:

t1 12:20 PM	HK1 0.02
	HK2 0.08
T2 1:20 PM	HK1 0.04
	HK2 0.11
T3 2:20 PM	HK1 0.08

		HK2 0.17
	T4 3:20 PM	HK1 0.21
		HK2 0.5
	T5 3:55 PM	HK1 0.5
		/

- Cultivation aborted at OD=0.5
- Cell suspension filled in 50 mL Falcons
- Centrifuged at 5000 rpm & 4 °C for 10 min
- Supernatant discarded
- Resuspended in 15 mL of ice-cold $CaCl_2$
- Centrifuged again
- Resuspended again
- Centrifuged again
- With 2.5 mL of $CaCl_2$+15% glycerol resuspended
- Aliquoted to 200 µL & 1x 50 µL in Eppis
- In N_2shock-frozen
- Frozen at -70 °C

Primer SB1 to SB6 suspended with ddH20 as indicated on the delivery note, diluted 1:10 to 10µM

Date	11.10.2024
Who	Noemi
What	Competent E. coliα λ pir manufactured: • OD the pre-culture: VK1: 10; VK2: 9 • Main culture 1&2 inoculated at 11:35 AM • Inoculated main culture 2 additionally with 1.5 mL of the pre-culture at OD 0.1 at 12 o'clock • OD measurement of the main cultures:

t1 12:20 PM	HK1 0.02
	HK2 0.08
T2 1:20 PM	HK1 0.04
	HK2 0.11
T3 2:20 PM	HK1 0.08
	HK2 0.17
T4 3:20 PM	HK1 0.21
	HK2 0.5
3:55 PM	HK1 0.5
	/

- Cultivation aborted at OD=0.5
- Cell suspension filled in 50 mL Falcons
- Centrifuged at 5000 rpm & 4 °C for 10 min
- Brushed off
- Resuspended in 15 mL of ice-cold $CaCl_2$
- Centrifuged again
- Resuspended again
- Centrifuged again
- With 2.5 mL of $CaCl_2$+15% glycerol resuspended
- Aliquoted to 200 µL & 1x 50 µL in Eppis
- In N_2shock-frozen
- Frozen at -70 °C

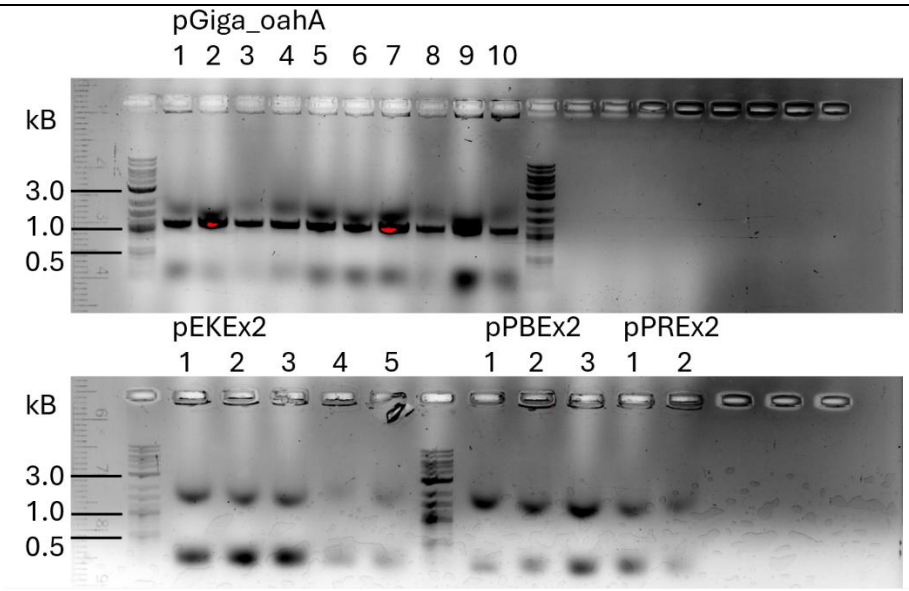
	Primer SB1 to SB6 suspended with ddH20 as indicated on the delivery note, diluted 1:10 to 10µM

Date	14.10.2024
Who	Ayleen
What	Transformed colonies from 11.10.24 transferred to grid plate

Date	15.10.2024
Who	Alex
What	Colony PCR • One Taq-protocol used • 2 mastermixes: 1x for pGgia_oahA, 1x for remaining plasmids: • 25 µL aliquots prepared in PCR tubes • Colony material added • PCR started ○ pGiga Annealing Temp: 51 °C; Elongation Time: 69 s (Fragment: 1138 bp) ○ pEKEx2 etc.: Annealing Temp 47 °C; Elongation time 12 s (~180 bp?)

	pGiga_oahA [µL]	pEKEx2/pPBEx2/pPREx2 [µL]
OneTaq Mastermix with Std Buffer	150	425
Forward Primer	6 (SB1_pGiga_oahA_for)	17 (SB5_CG_seq_oahA_for)
Reverse Primer	6 (SB2_pGiga_oahA_rev)	17 (SB6_CG_seq_oahA_rev)
Water	138	391

Date	16.10.2024
Who	Alex
What	Gel electrophoresis of the colony PCR: • 500 mL 1% agarose prepared: ○ 5 g agarose filled up to 500 mL with 1x TAE buffer ○ Heat in microwave until clear solution • Add 5 µL loading dye to the PCR samples, run the gel for about 30 minutes at 96 V.

	 With positive clones overnight cultures set up for cryo production: • pGiga_oahA: No 3 • pPBEx2: No. 1 • pEKEx2: No. 1 • pPREx2: No. 1 • 4x 10 mL LB medium in 100 mL shaking flask • + 10 µL Amp100 for pGiga_oahA • +5 µL Kan50 for the rest • Inoculated with the respective colony • Incubated at 37 °C with shaking

Date	17.10.2024
Who	Alex
What	Production of Cryos: • 12 cryo tubes labeled • 250 µL sterile 87%-glycerol presented in cryo-tubes • 750 µL of the overnight cultures added • Mixed • Frozen at -70 °C

Date	04.11.2024
Who	Ayleen
What	Overnight cultures for plasmid isolation of pGiga_oahA, pEKEx2, pPREx2, and pPBEx2 • 2x 500 mL LB medium prepared and autoclaved • Each 10 mL LB in 100 mL flask + 10 µL Amp100 or Kan50 • Inoculated with colony from plate at 3:30 PM • Incubated at 37 °C overnight

Date	05.11.2024
Who	Noemi

What	Plasmid isolation pGiga_oahA, pEKEx2, pPREx2 and pPBEx2 • SOP see Plasmidiso kit • OD measurement of the ON cultures and used volumes: ○ pGiga_oahA: 2.6→6.2 mL ○ pEKEx2: 3.6→4.4 mL ○ pPREx2: 4 →4 mL ○ pPBEx2: 3.8→4.2 mL • required quantity in Eppi pelletized • Rest see SOP • Concentrations:

Plasmid	[ng/µL]	A260/230	A260/280
pGiga_oahA	47,8	1,74	2,05
pEKEx2	518,2	2,18	1,90
pPREx2	284,4	2,09	2,00
pPBEx2	669,1	2,15	1,92

• Plasmids frozen at -21 °C

Date	06.11.2024
Who	Alex, Noemi
What	Q5-PCR: Production of the oahA fragment from pGiga_oahA • 3x 50 µL aliquots→see SOP • 2 µL pGiga used in 1:10 dilution (only 10 ng required) Restriction digestion of pEKEx2 and pPBEx2→Linearization for Gibson Assembly • 3 reactions per vector • PstI-HF and SalI-HF • Reaction:

	pEKEx2	pPBEx2 (2x)
DNA	1.93 µL	1.5 µL
10x NEB Buffer	5 µL	5 µL
enzyme	1 + 1 µL	1 + 1 µL
H_2O	41.1 µL	42.5 µL

• Incubated at 37 °C for 5-15 min
• heat inactivated at 65 °C for 20 min
• Checked with gel electrophoresis

gel electrophoresis
• 100 mL 1% agarose + 5 µL PEQGreen
• 30 min runtime at 96 V
• 20 µL of sample + 5 µL Purple Loading Dye
• Scheme:

1	4	7	10	13
Purple 1 kB Plus Ladder	oahA	pPBEx2 #1	pPBEx2 #2	pEKEx2

→good results

PCR clean-up
• See NucleoSpin Kit SOP
• Bottles NT3 buffer used in the 1st wash cycle
• Washed 2 more times with the right buffer

- Eluted with 30 µL NE
- Concentration measurement with NanoDrop:

oahA	4.801 ng/µL
pPBEx2 #1	7.063 ng/µL
pPBEx2 #2	5.268 ng/µL
pEKEx2	2.412 ng/µL

  - →far too low concentrations; new Q5-PCR and restriction digest

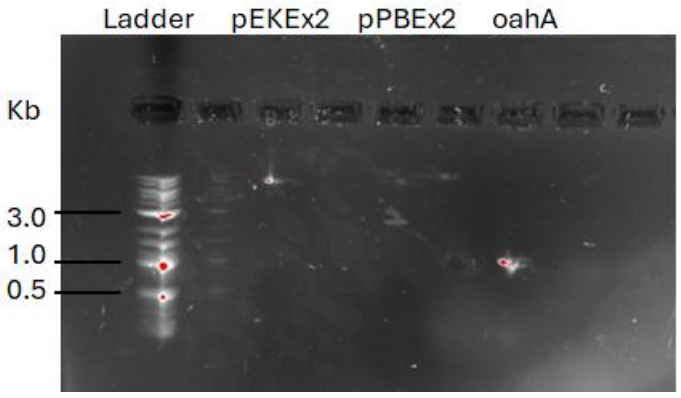
New Q5-PCR & restriction digestion

- Just like above
- All samples frozen at -21 °C

Date	07.11.2024
Who	Daniel, Yazan, Alex, Noemi
What	Gel electrophoresis of the PCR and restriction products from 06.11.25

- 20 µL sample + 5 µL loading dye
- Scheme:

1	3	5	7
Ladder	pEKEx2	pPBEx2	oahA



Cleanup of the PCR and restriction reactions

- See NucleoSpin Kit SOP
- Concentration measurement with NanoDrop:

pEKEx2	63.858 ng/µL
pPBEx2	77.614 ng/µL
oahA	508.414 ng/µL

Gibson assembly pEKEx2_oahA and pPBEx2_oahA:

- See NEBuilder protocol

	pEKEx2_oahA	**pPBEx2_oahA**
Vector	2 µL pEKEx2	2 µL pPBEx2
oahA 1:2 diluted	1 µL	1 µL
water	7 µL	7 µL
MasterMix	10 µL	10 µL

- 50 °C, 15 min

Date	08.11.2024
Who	Noemi
What	Heat shock of the Gibson Assembly products from 7.11.24 in E. coli • See protocol from 11.10.24 • 4 approaches: ○ 2 µL pEKEx2_oahA into 50 µL chemically competent E. coli ○ 2 µL pPBEx2_oahA + 50 µL E. coli ○ 4 µL pEKEx2_oahA + 50 µL E. coli ○ 4 µL pPBEx2_oahA + 50 µL E. coli • Plated each 100 µL and spin down on LB-Kan50 • Incubated over the weekend at RT

Date	11.11.2024
Who	Ayleen, Noemi
What	New heat shock of the Gibson assembly products from 11/7/24 in E.coli, as too few colonies grew over the weekend. • 2 reactions: ○ 10 µL pEKEx2_oahA + 50 µL E. coli ○ 10 µL pPBEx2_oahA + 50 µL E. coli • Rest see 11.10.24 • New plates and plates incubated overnight at 37 °C on 08.11.24

Date	12.11.2024
Who	Alex, Ayleen
What	*Even with the plates from 11.11, very few colonies grew; therefore, the colonies from 08.11.24 were examined by Colony PCR.* Colony PCR: • One Taq protocol; 25 µL reactions; 12 tests

OneTaq Master Mix	150 µL
10 µM Forward Primer	6 µL
10 µM Reverse Primer	6 µL
Water	138 µL

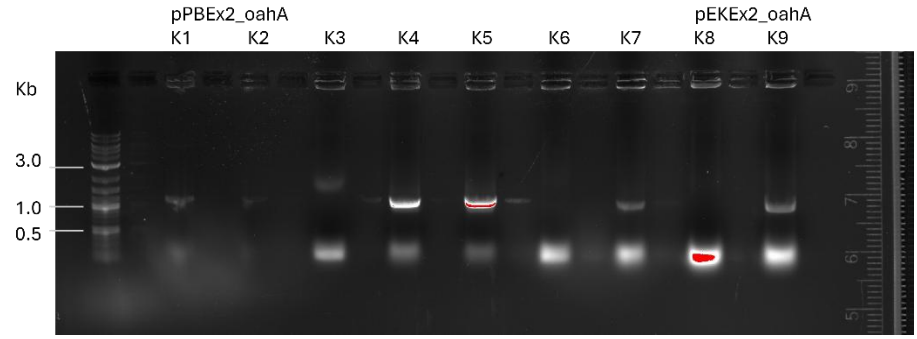
Remaining colonies from 11.11.24 striked onto grid plate

Date	13.11.2024
Who	Ayleen, Alex
What	AGE of the colony PCR products from 12.11.24 • See protocol 6.11.24 • 20 µL sample + 5 µL Loading Dye • Loading scheme:

1	3	5	7	9	11	13	15	17	19
1 kB Plus Ladder	K1	K2	K3	K4	K5	K6	K7	K8	K9

• K1 to K7: pPBEx2_oahA

- K8 & K9: pEKEx2_oahA



→K4, K5, K9 = positive

Overnight culture of the positive colonies:

- S. 4.11.24
- Inoculated at 1:45 PM
- Incubated at 37 °C

Date	14.11.2024
Who	Nio, Noemi
What	Cryos and plasmid isolation of overnight cultures

Used volumes:

	OD600	Plasmid prep	Cryo
K9	3,9	4 mL	5 mL
K5	4,2	4 mL	5 mL
K4	3,3	5 mL	4 mL

Cryos:

- Tubes labeled (Colony, Date, E. coli)
- 250 µL sterile 87%-glycerol presented in tubes
- Add 750 µL of the cell suspension and mix by pipetting.
- Stored at -70 °C

Plasmidiso of pEKEx2_oahA and pPBEx2_oahA:

- Required volumes in 15 mL Falcon tubes pelleted by centrifugation
- Rest see 'Nucleo Spin Plasmid Easy Pure' Kit SOP
- Incubated for about 3 minutes with elution buffer (actually too long, but no bad results)
- Concentration measurement with NanoDrop:

K5 (pPBEx2_oahA)	616,999 ng/µL
K4 (pPBEx2_oahA)	709,102 ng/µL
K9 (pEKEx2_oahA)	366,175 ng/µL

- Plasmids frozen at -21 °C

Media prepared for electroporation:

- See Marvin's SOP
- BHI (Wear a mask due to fine dust!)

	• BHI+sorbitol • BHI-Agar • HEPES 0.5 M, 100 mL • EPBI 0.5 L • EPBII 0.2 L →everything still has to be autoclaved and sorbitol added to 'BHI+sorbitol'

Date	15.11.2024
Who	Noemi
What	Media from 14.11.24 autoclaved Q5-PCR to check whether isolated plasmids (from K4, K5, K9) contain the correct fragment: • 2x 50 µL reaction per plasmid • Reactions: (see also SOP)

5x Q5 reaction buffer	10 µL
10 mM dnTPs	1 µL
10 µM Forward Primer	2.5 µL (SB1)
10 µM Reverse Primer	2.5 µL (SB2)
Template DNA	2 µL plasmid
Q5 High Fidelity DANN-Polymerase	0.5 µL
Nuclease Free Water	31.5 µL

Note: Template DNA should have a maximum of 5 ng/µL, therefore it was diluted as follows:

- K5 - 1:150 = 4.11 ng/µL (1 µL K5 + 149 µL H2O)
- K4 – 1:150 = 4.67 ng/µL (see above)
- K9 – 1:100 = 3.66 ng/µL (1 µL K9 + 99 µL H2O)
- PCR settings see Marvin's SOP

BHI agar plates:

- 0.5 L BHI-agar supplemented with 500 µL Kan50
- 18 plates cast (BHI Kan50)

1% agarose prepared:

- Filled 5 g of agarose with 1x TAE buffer to 500 mL
- Heat in microwave until clear solution

AGE of the PCR products:

- 100 mL gel poured into large sled (cf. 6.11.24)
- Loading scheme:

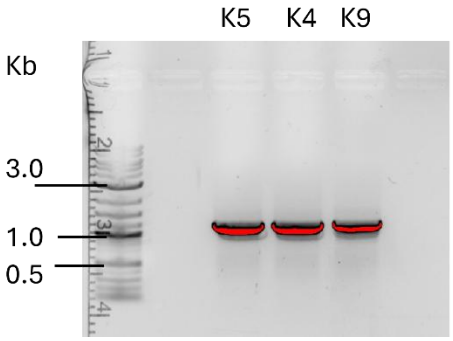
1	2	3	4	5	6	7	8	9
1 kB Plus Ladder	K5	K5	-	K4	K4	-	K9	K9

-> no positive results, image not saved (no bands visible)

New PCR; this time OneTaq instead of Q5:

- Mastermix prepared for 6x 50 µL reaction:

10 µM Forward Primer	6 µL (SB5)

	10 µM Reverse Primer: 6 µL (SB6) OneTaq MM with Std Buffer: 150 µL Water: 126 µL • Aliquoted into PCR tubes of 48 µL • 2 µL template DNA added • PCR setting see Marvin's script AGE of the OneTaq products • 40 mL gel poured into small sleds (40 mL 1% agarose + 2µL PEQGreen) • Load: 1: Ladder; 2: ; 3: K5; 4: K4; 5: K9 correct lengths in all samples:  C. glut WT spread on BHI; incubated over the weekend at room temperature

Date	18.11.2024
Who	Yazan, Alex
What	Sequencing of the plasmids from K5 (pPBEx2_oahA) and K9 (pEKEx2_oahA): • 5 µL DNA + 5 µL plasmid in sequencing tubes→desired concentrationssee sequencing box instructions • Plasmid diluted 1:10, Primer 1:2 • Tube Occupancy: pPBEx2_oahA (K5) / SB5 / EF73137798 pPBEx2_oahA (K5) / SB6 / EF73137799 pEKEx2_oahA (K9) / SB5 / EF73137800 pEKEx2_oahA (K9) / SB6 / EF73137801 Pre-cultures for electroporation: • 2x 5 mL BHI medium inoculated with C. glut WT colony in glass tubes • Incubated overnight at 30 °C and 140 rpm

Date	19.11.2024
Who	Nio, Noemi
What	Electroporation of WT with pEKEx2_oahA and pPBEx2_oahA: • See Marvin's SOP

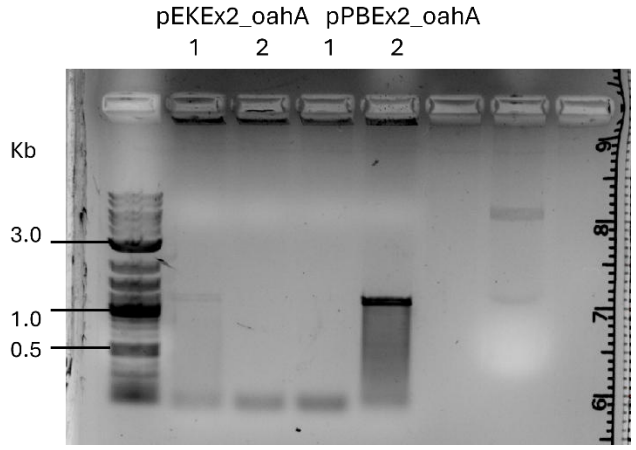
	• 100 mL of sorbitol added to 'BHI+sorbitol'*(first too little; cells only grew on BHI+10 mL sorbitol)* • 2 main cultures inoculated at 10:10 AM • After 1 h: OD600 = 0.6 • Stored on ice for 30 minutes, as ampicillin was not prepared. • incubated for 1 hour with ampicillin • Rest see SOP; possibly plasmids swapped • ÜN culture vaccinated & incubated BHI agar plates & BHI+Kann25 agar plates • See Marvin's SOP • Added 250 µL Kan50 to BHI+Kan25-Medium after autoclaving C.g. WT Cryos • see 17.10. • 12 aliquots frozen at -70 °C BHI+Sorbitol corrected with missing 68,5 mL sorbitol

Date	20.11.2024
Who	Ayleen, Noemi
What	Plating of electroporated cultures • Cultures centrifuged at 12000g, supernatant discarded, pellet resuspendet • Plated on BHI-Kan25 plates

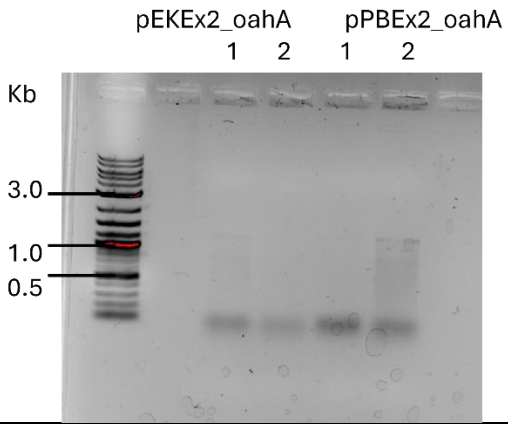
Date	22.11.2024
Who	Daniel, Noemi
What	Bacterial lawn on both plates from 20.11. Therefore: re-plate onto different plates to test whether our plates are defective / to isolate positive colonies • Kann25 BHI plates from Marvin ○ C. g. pEKEx2_oahA o C. g. pPBEx2_oahA • Kann25 BHI plates from SynBio: ○ C. g. WT ○ C. g. pEKEx2_oahA • Kann50 BHI plates from SynBio: ○ C. g. WT ○ C. g. pPBEx2_oahA

Date	25.11.2024
Who	Alex, Noemi
What	Colony PCR of elektroporated cells from 22.11. • see OneTaq protocoll • 4x 50 µL Reaction AGE: • 40 mL 1% Agarose + 2 µL PEQGreen

1	2	3	4	5

	<table><tr><td>Ladder 1 kB Plus</td><td>pEKEx2_oahA 1</td><td>pEKEx2_oahA 2</td><td>pPBEx2_oahA 1</td><td>pPBEx2_oahA 2</td></tr></table> • Right band only in lane 5  Overnight culture for new electroporation • 1 × 5 mL BHI + WT colony in a glass tube • Incubated at 30 °C, with shaking Diluting colonies • Each of *C. g. pEKEx2_oahA* and *C. g. pPBEx2_oahA* streaked out on new Kann25 BHI plates using the three-line streak method • Incubated at 30 °C

Date	26.11.2024
Who	Nio, Noemi
What	Elektroporation of WT with pEKEx2_oahA und pPBEx2_oahA: • see Marvins SOP (→ methods) • 3 cultures inoculated: 2 for elektroporation, 1 for OD-measurement at 9:45 AM • Incubated at 30 °C under shaking conditions • OD = 0,76 reached at 12:40 Uhr (slightly too high, but we continued nevertheless) • Rest see SOP Colony-PCR of diluted colonies from 25.11.24 • OneTaq protocol • 2x amount made, half amount freezed as stock for later use AGE of Colony-PCR reactios • Ladder; -; pEKEx2_oahA#1; pEKEx2_oahA#2; pPBEx2_oahA #1; pPBEx2_oahA #2 • results: only slight band: pPBEx2_oahA #2 zwischen 1.0 und 1.2 kB



Date	27.11.2024
Who	Alex, Ayleen
What	Plating of electroporated cultures: • centrifuged in eppis • supernatant discarded, pellet resuspended and plated on BHI-Kan25 • incubated at 30 °C

Date	28.11.2024
Who	Noemi
What	2 colonies of each plate from 27.11.2024 streaked on BHI+Kan25

Date	29.11.2024
Who	Alex, Nio
What	Colonies from 28.11. plated on grid with small pipette tips

Date	02.12.2024
Who	Nio, Noemi
What	Colony PCR of the colonies from grid plates • 4 clones of each plate tested + 2 comparisons with freezed Mastermix (to see if mastermix can be stored frozen) • 17x 50 µL = 850 µL reaction mastermix made and aliquoted in pcr tubes:: • PCR tubes labeling: • Thermocycler-settings see SOP

10 µM Forward Primer (SB5)	17 µL
10 µM Reverse Primer (SB6)	17 µL
OneTaq Quick Load Mastermix	425 µL
Water	391 µL

1.1 – 1.4	clones 1 to 4 pPBEx2_oahA_1
2.1 – 2.4	clones 1 to 4 pPBEx2_oahA_2
3.1 – 3.4	-„“- pEKEx2_oahA_1
4.1 – 4.4	-„“- pEKEx2_oahA_2
5.1	Colony 1 pPBEx2_oahA_1 with frozen mastermix 50 µL
5.2	-- 25 µL (not enough for more)

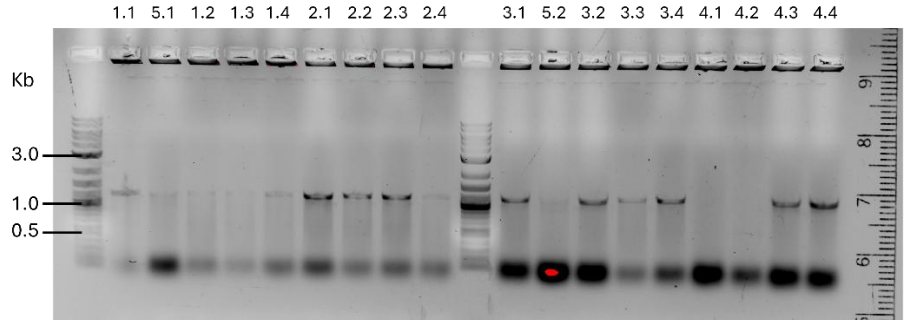
AGE of PCR-products

- big gel poured: 80 mL 1% Agarose + 4 µL PEQGreen
- loading scheme:

1	2	3	4 to 6	7 to 10
Ladder	1.1	5.1	1.2 – 1.4	2.1 – 2.4

11	12	13	14 to 16	17 to 20
Ladder	3.1	5.2	3.2 – 3.4	4.1 – 4.4

- Positive clones: 2.1, 2.2, 2.3, 3.1, 3.2, 3.3, 3.4, 4.3, 4.4



Date	3.12.2024
Who	Nio
What	ON-cultures of positive clones (4.3 and 2.2 see 02.12.24) • 2x 10 mL BHI+Kan25 and 2x 10 mL BHI+Kan50 in 100 mL shake flask inoculated with a single colony ~11:30 AM • Incubated at 30 °C and 120 rpm over night

Date	04.12.2024
Who	Nio, Noemi
What	Cryos of positive clones • see 14.11.24 OD-measurement of overnight cultures:

culture	OD_{600}
Kan25 pPBEx2_oahA	15,436
Kan50 pPBEx2_oahA	15,32
Kan25 pEKEx2_oahA	16,44
Kan50 pEKEx2_oahA	14,50

Date	09.12.2024
Who	Nio, Noemi
What	CGXII-stock • CaCl2-, MgSO4-Lösung in falcons filter-sterilized • 1 mL CaCl2, MgSO4 and Biotin added to CGXII-stock (sterile) Precultures for first cultivation

	• 10 mL CGXII-Medium added to 3x 100 mL shake flasks (9.2 mL stock, 0.8 mL 50%-Glucose, 10 µL PCA, 10 µL SEL) • 10 µL Kan50 added to 2 flasks (for cultures with plasmid) • inoculated with C.g. wildtype, pEKEx2_oahA and pPBEx2_oahA • incubated at 30 °C under shaking conditions over night

Date	10.12.2024
Who	Noemi
What	Second preculture • 3x 250 mL shake flasks filled with 25 mL CGXII-medium • each flask inoculated with 125 µL of second preculture • 25 µL Kan50 added to cultures with plasmid OD-measurement of first preculture:

culture	dilution	OD_{600}
wildtype	1:1000	0.0194
pEKEx2_oahA	1:1000	0.0211
pPBEx2_oahA	1:1000	0.0659

flasks for main culture prepared: 5x 500 mL shake flasks with screw caps and needles for sampling autoclaved

Date	11.12.2024
Who	Noemi, Nio
What	Main cultures • Prepared shake flasks filled with 50 mL CGXII-medium each • OD-measurement of 2.preculture and determining inoculation volume for OD=0.5 (v1*c1=v2*c2):

culture	OD_{600}	$V_{inoculum}$ [µL]
WT	60,32	374
pPBEx2_oahA	64,25	384
pEKEx2_oahA	60,36	410

• inoculation ~10:50 AM
• incubation ~11:00 AM at 30 °C under shaking conditions (120 rpm)
• Eppis for sampling labeled: time (t0 etc.), culture (WT,E-,E+,B-,B+), type (pellet or supernatant)
• Every 1.5 h ~1.5 mL sample taken, OD measured, centrifuged (12000 rpm, 4 min), pellet and supernatant stored seperately at -20 °C
• At 02:20 PM 2.5 mM IPTG added to pPBEx2_oahA+ and pEKEx2_oahA+

IPTG-solution prepared
• 2.68 g IPTG dissolved in 10 mL ddH_2O and filter-sterilized -> 100x stock for 2.5 mM end concentration in 45 mL-cultures

OD-measurement: (11. – 12.12.2024)

sample	time	dilution	OD_{600}				
			WT	pEKEx2	pEKEx2 +	pPBEx2	pPBEx2 +
t0	11:00	-	0,5	0,5	0,5	0,5	0,5
t1	12:30	1:2	0,4981	0,5454	0,5749	0,5238	0,5133
t2	14:00	1:20	0,0833	0,0952	0,0916	0,0863	0,0981

	t3	15:30	1:20	0,1683	0,1441	0,1393	0,1465	0,1293
	t4	17:00	1:20	0,3302	0,3198	0,3044	0,2962	0,273
	t5	18:00	1:100	0,1081	0,1034	0,0681	0,0852	0,0775
	t6	09:00	1:200	0,3656	0,3311	0,2556	0,2855	0,3235
	t7	10:30	1:200	0,2481	0,2969	0,2996	0,3115	0,3016
	t8	12:00	1:200	0,249	0,2804	0,279	0,2761	0,3048
	t9	13:30	1:200	0,2428	0,3139	0,2924	0,2938	0,2941
	t10	15:40	1:200	0,2511	0,3405	0,2677	0,2769	0,3214

Date	12.12.2024
Who	Noemi
What	Main culture continuation First oxalate analysis using colorimetric Oxalate Assay Kit (Sigma-Aldrich) • See SOP • OD595-data:

sample	dilution	OD595
WT Internal standard	-	0,2022
WT Sample	-	0,0262
pPBEx2_oahA+ IS	1:200	0,0476
pPBEx2_oahA+ Sample	1:200	0,0341

• Oxalic acid-concentration: [Ox] = $\frac{OD_{sample}\text{-}OD_{Blank}}{OD_{IS}\text{-}OD_{Sample}} * 500 * df\ [\mu M]$

• → WT: = 74,43 µM

• → pPBEx2_oahA+: =359,48 µM

Date	13.01.2025
Who	Alex, Ayleen
What	First Preculture for induction strentgh test • See 09.12.2024 • Medium turned light violett

Date	14.01.2025
Who	Ayleen
What	Second Preculture (see 10.12.2024)

Date	15.01.2025
Who	Noemi
What	Main cultures: 10x 50 mL CGXII-medium with Kan50 where necessary 1. Cg WT 2. pPBEx2_oahA 3. pPBEx2_oahA 0.5 mM IPTG 4. pPBEx2_oahA 1.0 mM IPTG 5. pPBEx2_oahA 1.5 mM IPTG 6. pPBEx2_oahA 2.0 mM IPTG 7. pPBEx2_oahA 2.5 mM IPTG

	8. pPBEx2_oahA 3.0 mM IPTG 9. pPBEx2_oahA 5.0 mM IPTG 10. pPBEx2_oahA 2.5 mM IPTG at t0 • cultures were inocultated to OD=0.5 at 11:50 AM • IPTG was added to cultures 2-9 at 04:20 PM at indicated levels • For growth curve see Results • OD was measured every 1.5 h, samples were frozen at -21 °C Andrew Pipetting Robot started to measure oxalate concentration in samples of first cultivation (see Excel document)

Date	16.01.2025
Who	Nio
What	Cultivation Day 2 Measurement continued as before Cultivation aborted at 04:00 PM 3 tubes per shaking flask and time measured OD of 0.6 ml each centrifuge rest

Date	21.01.2025
Who	Noemi
What	ON-Culture for plasmid isolation and sequencing 2 x 100 ml shaking flask 10 ml BHI-media each 10 µl Kan50 to each Inoculate with *c.g. pEKEx2*_oahA and *c.g. pPBEx2_oahA*

Date	22.01.2025
Who	Alex
What	Measured absorbance according to the SOP of the Merck MAK315-1KT Oxalate Assay Kit, except for step 2 Plasmid isolation of c.g. pPBEx2_oahA (Kan50) & c.g. pEKEx2_oahA (Kan50) OD of precultures: pEKEx2_oahA (dil. 1:100) 0.1382 pPBEx2_oahA (dil. 1:100) 0.1795 1 ml of pEKEx2_oahA & 660 µl of pPBEx2_oahA used Isolation according to SOP NucleoSpin Kit for gram-positive bacteria 25 µl of 25 mg/ml lysozyme solution per tube Incubated for 30 minutes at 37°C Nanodrop-Measurements: pEKEx2_oahA 27.072 ng/µl both very impure pPBEx2_oahA 10.726 ng/µl → sent *E. col*i plasmid isolation products K5, K9 for sequencing → sent Q5 Colony PCR product of the *c.g.* precultures from 21.01.2025 for sequencing

	Q5 Colony PCR: • Program: • 54°C annealing temperature • Used 2 µl of cell suspension • Procedure according to SOP Sequencing of pEKEx2_oahA (K9) and pPBEx2_oahA (K4, K5) • K9: diluted 1:5 (10 µl plasmid + 40 µl H2O) • K4: diluted 1:10 (5 µl plasmid + 45 µl H2O) • K5: diluted 1:10 (5 µl plasmid + 45 µl H2O) • SB5 primer diluted 1:2 • Mixed 6 µl pasmid with 5.48 µl primer → transferred into 10 µl tube

Date	23.01.2025
Who	Noemi
What	PCR-cleanup of products from 22.01.2025 with NucleoSpin Kit • Sample/buffer NTI = 1/2 → 50 µl fragments + 100 µl NTI • Rest according to SOP • Eluated with 30 ml NE buffer Nano-Drop measurements:

Sample:	concentration	A260/230	A260/280
pPBEx2_oahA	60.411 ng/µl	0.287	1.828
pEKEx2_oahA	76.310 ng/µl	0825	1.885

→ bad results

Repetition of cleanup; wash twice + eluate with 15 µl NE

Sample:	concentration	A260/230	A260/280
pPBEx2_oahA	64.618 ng/µl	1.756	1.867
pEKEx2_oahA	105.138 ng/µl	1.558	1.908

→ eluate with water next time

Sequencing:
- 10 ng/µl fragment, 5 µl 2x
- 5 µM primer, 5µl 2x
- → dilute oahA from pPBEx2 1:12 → 15 µl PCR product + 165 µl ddH2O
- → dilute oahA from pEBEx2 1:20 → 15 µl PCR product + 285 µl ddH2O
- → dilute primer 1:2 → 15 µl primer + 15 µl ddH2O

Date	27.01.2025
Who	Ayleen
What	Inoculated 3 x 10 ml precultures (see 09.12.2024)

Date	28.01.2025
Who	Alex
What	Preparation of the FlowerPlate for the Biolector 20 µl IPTG solution per well → triplicates for each IPTG concentration and vector Prepare 7*20 ml = 140 µl solution for each IPTG concentration

Sample concentration	IBTG (µl)	ddH2O (µl)
0.5 mM	3.12	136.88
1.0 mM	6.23	133.77
1.5 mM	9.35	130.65
2.0 mM	12.46	127.54
2.5 mM	15.58	124.42
3.0 mM	18.69	121.31
5.0 mM	31.15	108.85

Preparation of media:

- 1 ml per well → 48 ml media → preparation of 50 ml media
- 44 ml CGXII
- 4 ml Glucose solution
- 50 µl PCA
- 50 µl SEL
- 50 µl Ka50

OD measurement of precultures:

- Dilution by 1009:10

WT: 0.31	pPBEx2: 0.50	pEKEx2: 0.47

- →preparation of a cell suspension of 480 µl total volume

Sample:	Cell suspension (µl)	ddH2O (µl)
WT	15.985	569.015
pPBEx2	9.911	470.015
pEKEx2	10.543	469.457

- → preparation of a master mix

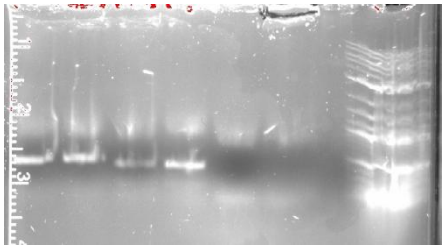
Sample:	Cell suspension (µl)	ddH2O (µl)
WT	64	1856
pPBEx2	228	20342
pEKEx2	232	20328

Date	30.01.2025
Who	Ayleen
What	Biolector samples transfered to 1.5 ml tubes (1 ml) → centrifuged Supernatant transferred into new 1.5 ml tubes Transferred growth curves to USB-Stick

Date	03.02.2025
Who	Daniel, Yazan
What	Prepared precultures: • *C. glutamicum* WT (CGXII media) 10 ml • *C. glutamicum* pEKEx2 (CGXII media; Kan50) 10 ml • *E. coli* DH5αλpir pKK19 (LB media; Kan50 (10ml); inoculated with 100 µl cell suspension) 10 ml • Volumes see 21.01.2025

Date	04.02.2025
Who	Daniel, Noemi

What	Plasmidisolation (NucleoSpin) (SOP see 14.11.2025) → plasmid concentration: 71 ng/µl Restriction digestion (EcoRI + BamHI) (SOP see 06.11.2024) • 23.5 µl plasmid • 30 µl ddH2O • 5 µl CutSmart buffer • 2 µl EcoRI-HF • 2 µl BamHI-HF Dephosphorylation (→ see Methods) Making fragments: • Fragment 2: (tac + lac operon + oahA) from PEKEx2 *C. glutamicum* • Fragment 3 & 1: homologous flanks from *C. glutamicum* WT

Date	05.02.2025
Who	Daniel
What	Gel electrophoresis not successful (01.02.2025) → repetition needed 

Date	11.02.2025
Who	Noemi, Daniel
What	New Q5 PCT for 3 fragments: • oahA + tac + lac promoter from pEKEx2_oahA "K9" → primer annealing at 71°C • 2 homologous sequences from C. glutamicum WT → primer annealing at 54°C Prepared AGE Preculture of *E. coli* DH5α in 10 ml preculture at 37°C

Date	12.02.2025
Who	Noemi, Daniel
What	Prepared new GE gel • 40 ml 1% agaorose + 2 µl PeqGreen • Had to be redone due to air bubbles in gel Sample preparation: • Mixed all fragments with 10 µl loading dye • 2 x 25 µl of each sample in pockets • Loading scheme: -|1kb+ ladder| F1|F1|F2|F2|F3|F3 Plasmid isolation pk19:

- Nucleospin; eluated with 30; 4 batches
- Concentrations:

Sample	1	2	3	4
c ng/µl	16	74	27	64

4 Batches for restriction digestion:

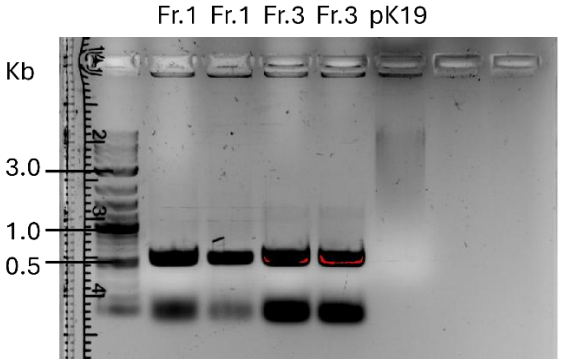
- 30 µl plasmid
- 5 µl buffer
- 2.5 µl BamHI
- 2.5 µl EcoHI
- 10 µl ddH2O
- → incubate for 30 minutes at 37°C

After 30 minutes added phosphatase and buffer → incubate for 30 minutes at 37°C
→ 2 minutes at 80°C for stopping digestion

New PCR for making fragments 1 & 3 (homologous sequences):

- 2x reaction:
- 5x Q5 Reaction buffer: 10 ml
- 10 mM dNTPs: 2 µl
- 100 µM forward primer: 0.5 µl
- 100 µM reverse primer: 0.5 µl
- Template DNA: -
- Q5 High-frequency DNA-polymerase: 1 µl
- Nuclease-free water: 33 µl
- DMSO: 3µl

- Pick C. glutamicum WT colonies
- 1 minute of elongation time (instead of 16 seconds)

Date	13.02.2025
Who	Noemi
What	Gelelectrophoresis of fragments 1 & 3 and linearized pK19 • Loading scheme: 1 kB Plus Ladder | Fragment 1 | Fr.1 | Fr. 3 | Fr. 3 | pK19  Gel clean-up of Fragment 1 (Fr. 3 did not have the right length) • Nucleo Spin Gel&PCR clean-up Kit • Concentration: 37.8 ng/µL and 18.2 ng/µL → enough for assembly

Date	17.02.2025
Who	Daniel, Yazan
What	New Fragments produced with Q5-PCR for pK19_oahA assembly • Reaction composition see protocol • New primers used AGE with the PCR products & clean-up of the 3 fragments • Concentrations: 1 – 110 ng/µL | 2 – 106,7 ng/µL | 3 – 47,4 ng/µL Overnight culture for plasmid-prep: • E. coli_pK19 from plate in 10 mL LB-medium • Incubated at 37 °C under shaking conditions

Date	18.02.2025
Who	Yazan, Daniel
What	pK19 plasmid isolation • 2 preps: one with water elution, one with AE buffer elution • Concentrations: water-elution: 310 ng/µL | buffer-elution: 350 ng/µL pK19 restriction digest with BamHI and EcoRI for assembly

pK19	6 µL
BamHI	2 µL
EcoRI	2 µL
Cutsmart Buffer	5 µL
ddH_2O	35 L

• Incubated for 60 min at 37 °C
• After 30 min: addition of 5 µL antarctic phosphatase buffer & 2.5 µL antarctic phosphatase
• After 60 min: heat inactivation → 2 min 80 °C

AGE of restriction digest +clean-up (see Kit SOP)
• Concentrations: lin. pK19_1 – 23.4 ng/µL | lin. pK19_2 – 17.6 ng/µL | lin. pK19_3 – 19.2 ng/µL

Date	19.02.2025
Who	Yazan, Noemi
What	Gibson Assembly Fragment 1 Dilute 1:1 1 µL Fragment 2 1 µL Fragment 3 1 µL Transformation in *E. coli* 100 µL competent cells + 1 µL plasmid 20 min on ice 1 min at 42 °C 1 min on ice 800 µL LB medium Incubate for 1h at 37 °C while shaking Spread on LB-Kann50 (100 µL spin-down) -> incubate overnight at 37 °C PCR of the Gibson Assembly product: Q5 PCR

	• ~2800 bp 86 sec, T= 58 °C annealing,ΔT -> -0.1 °C per cycle • SecP1 + SecP2 (primer from Marvin generally for pK19 •

Reaction Buffer	10 µL
10 mM dNTPs	1 µL
10 µM forward primer	2.5 µL
10 µM reverse primer	2.5 µL
Template DNA	8 µL
Q5 polymerase	0.5 µL
Nuclease-Free Water	25.5 µL

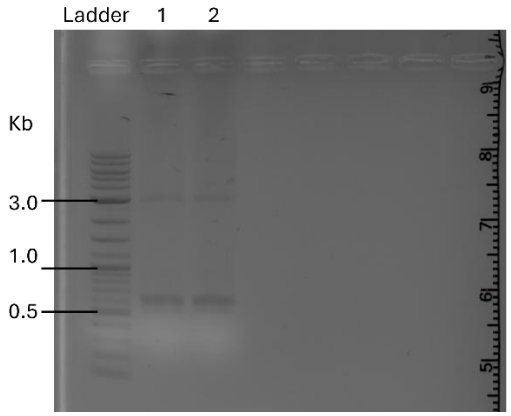
• Run overnight

Date	20.02.2025
Who	Noemi
What	AGE of the PCR product 40 mL 1% agarose + 2 µL PeqGreen 50 µL PCR product + 10 µL Purpleloading Dye 30 min running time

1	2	3
Purple 1kb Plus Ladder	PCR product 25 µL	PCR product 25 µL

Weak bands at ~3000 bp & ~550 bp

Gibson Assembly didn't work (plates didn't grow either)



New Gibson Assembly for pK19mobsacB_oahA_tac

- Lin vector (17.6 ng/µL) 6 µL
- Fragment 1 (110 ng/µL) 1 µL 1:2 diluted
- Fragment 2 (106.7 ng/µL) 1 µL
- Fragment 3 (47.4 ng/µL) 1 µL
- Assembly MM 10 µL
- Water 1 µL

Incubate for 1h at 50 °C

Transformer in E. coli

SOP see 19.02.2025, 2 batches: 100 µL compies + 10 µL DANN, 88 µL compies + 4 µL DNA

1h 10 min incubation time, plated on 4 plates (LB Kan50) & incubated overnight at 37 °C

PCR (Q5)

- Settings see 19.02.2025
- Reaction:

	Here: Dilute 1 µL of Gibson Assembly product with 3 µL of water & use 1 µL of it as template	Here: 5 µL pure Gibson Assembly product
Reaction Buffer	10 µL	10 µL
10 mM dNTPs	1 µL	1 µL
10 µM forward primer	2.5 µL	2.5 µL
10 µM reverse primer	2.5 µL	2.5 µL
Template DNA	1 µL	5 µL
Q5 polymerase	0.5 µL	0.5 µL
Nuclease-Free Water	32.5 µL	28.5 µL

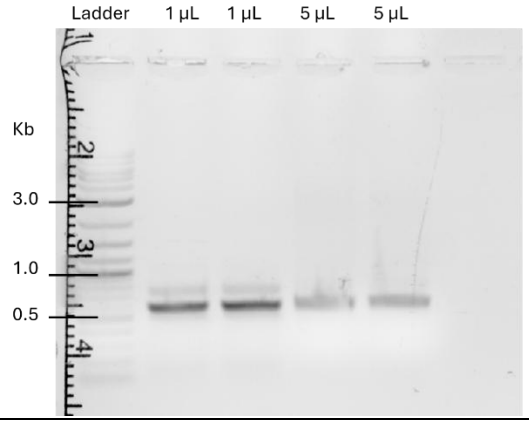
Date	21.02.2025
Who	Noemi
What	Plate restrictions Plate to grid: 14 colonies from “4 µL SD” plate 7 colonies from “10 µL SD” plate Left on the bench over the weekend AGE Load PCR products onto the gel 1 µL batch 5 µL batch

1	2	3	4	5
Purple 1kb Plus Ladder	1 µL	1 µL	5 µL	5 µL

10 µL loading dye per 50 µL PCR product

Let it run for 30 minutes

- Again only bands at ~550 bp, probably piece in pK19 vector between primers without insert

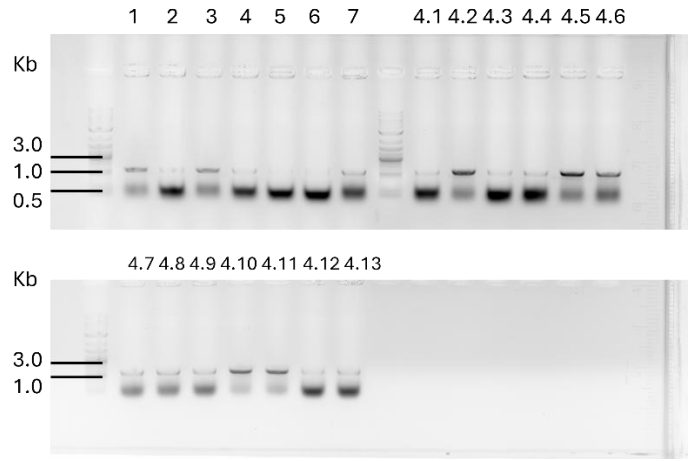
Date	25.02.2025
Who	Noemi
What	Colony PCR of transformed *E. coli* (pK19mobsacB_aohA_tac) One Taq protocol -> 25 µL reactions 20 batches (13 from 4 µL GA, 7 from 10 µL GA) MasterMix: 10 µM forward primer: 10.5 µL 10 µM reverse primer: 10.5 µL Template DNA: - OneTaq Quick Load MM: 262.5 µL Nuclease-Free Water: 241.5 µL Each colony was picked from the grid plate and transferred "1" to "7" = *E.coli*_pK19mobsacB_oahA_tac from 10 µL GA transformation "4.1" to "4.13" = *E.coli*_pK19mobsacB_oahA_tac from 4 µL GA transformation Annealing Temperature 52 °C

Date	04.03.2025
Who	Noemi
What	Pre-culture preparation E. coli KAN50 in LB medium see 12.02.2025

Date	07.03.2025
Who	Noemi
What	1% agarose See 15.11.2024 1 x TAE buffer filled 3 L: 60 mL 50 x buffer + 2940 mL deionized water AGE Pour 80 mL of 1% agarose + 4 µL of PeqGreen into a large slide PCR products from 25.02.2025 abandoned

30 min running time

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Lad-der	1	2	3	4	5	6	7	Lad-der	4.1	4.2	4.3	4.4	4.5	4.6
Lad-der	4.7	4.8	4.9	4.10	4.11	4.12	4.13							



Date	10.03.2025
Who	Noemi
What	Preculture for catalyst dust test 3 pre-cultures: Cg wild type Cg_pPBEx2_oahA x2 Prepare 3 x 10 mL CGXII medium: 3 x 9.2 mL CGXII stock 27.6 mL 3 x 0.8 mL Glc solution 2.4 mL 3 x 10 µL PCA 30 µL 3 x 10 µL SEL 30 µL For Cg_pPBEx2_oahA flask: 10 µL Kan50

Date	11.03.2025
Who	Noemi
What	New PCA manufactured (see media & stock solutions), ~2.5 mL 1 M NaOH added to dissolve Preculture for catalyst dust test • Prepare medium (for Wednesday already -> ~330 mL) ○ CGXII stock 322 ml ○ Glc solution 28 ml ○ PCA 350 µL ○ SEL 350 µL • 25 mL each in 3 x 250 baffled flasks • For Cg_pPBEx2_oahA, 25 µL each of Kan50 • Inoculation with 1st preculture with 1st preculture (400 µL)

	Bus catalyst solutions applied • 5% -> 2.5 g in 50 mL water • 20% -> 10 g in 50 mL water • 1% -> 0.5 g in 50 mL water -> more than 1% should probably not be used in cultures • pH values: ○ 5% ~ pH = 4 ○ 20% ~ pH = 3 to 4 ○ 1% ~ pH = 4.5 -> test tomorrow with CGXII (with CGXII stick + dust: pH = 7) Preculture for plasmid isolation • 3 x 10 mL LB medium in 100 mL flasks + 10 µL Kan50 inoculate with E. coli DH5α pK19 mobsacB

Date	12.03.2025
Who	Noemi, Janina, Julius, Amelie
What	Weigh 3 x 0.5 dust (sterile)

	Empty weight Eppis [g]	Final dust weight [g]
1	1.08	0.47
2	1.09	0.47
3	1.09	0.45

Prepare new CGXII 3 x 50 mL
138 mL CG XII Stock
12 mL 50% glucose
150 µL pca 11.03.2025
150 µL trace elements

OD 600 measurement

	OD600 [ ]	dilution
WT	0.4911	1:100
PP1	0.4294	1:100
PP2	0.4009	1:100

Adding road dust
WT: Dust 1
PP1: Dust 2
PP2: Dust 3

Required amount for OD 0.5 in main culture

$$V_1 = \frac{C_1 \cdot V_2}{\text{OD-Whot}}$$

- $WT = \frac{0{,}5 \cdot 50\ mL}{100 \cdot\ 0{,}4911} = 0{,}5091\ mL$
- $PP1 = \frac{0{,}5 \cdot 50\ mL}{100 \cdot\ 0{,}4294} = 0{,}5077\ mL$

- $PP2 = \frac{0{,}5 \cdot 50\ mL}{100 \cdot\ 0{,}4009} = 0{,}6236\ mL$

Inoculation of cultures: 12:00

Plasmid isolation

OD measurement:

OD = 3.1477

OD = 3.9793

Centrifuge

Rest see SOP

Elute twice with 30 µL AE

pK19 but eluted with 50 µL AE

	A260[ ]	*A260/230*[ ]	*A260/280*[ ]	*c*[ng/µL]
1e	3.7277	1.233	2.085	186.388
2e	8.7966	2.050	1.981	189.831
1s	4.0232	2.181	1.915	201.158
2s	4.6677	2.138	1.991	233.386

OD measurement main culture

1:30 p.m.	t1	WT	1:1	0.4300				
		pPBEx2	1:1	0.5116				
		pPBEx2+	1:1	0.5149				
3:00 PM	t2	WT	1:1	0.4814				
		pPBEx2	1:1	0.8264	1:2	0.5626	1:4	0.2189
		pPBEx2+	1:1	0.8047	1:2	0.6690	1:4	0.2702
4:30 p.m.	t3	WT	1:4	0.3595				
		pPBEx2	1:4	0.4581				
		pPBEx2+	1:4	0.5529	1:8	0.4019		
4:55 PM		Induced to 1 mM IPTG						
6:00 PM	t4	WT	1:8	0.4232				
		pPBEx2	1:8	0.4612				
		pPBEx2+	1:8	0.5136	1:6	0.2372		

- IPTG: c = 0.268 g/mL = 1.1246 M
  V = (1.5 mM * 45 mL)/1124.6 mM = 60 µL inoculation volume for 1.5 mM Final concentration in 45 mL culture

Restriction digestion

	EcoRI	BamHI
1	$V_{Plasmid}$= 1000 µg*µL / 201.2 µg = 5 µL	$V_{Plasmid}$ 5 µL
	V_{EcoRI}= 2 µL	V_{BamHI}= 2 µL
	V_{Buffer}= 5 µL	V_{Buffer}= 5 µL
	V_{H2O}= 38 µL	V_{H2O}= 38 µL
2	$V_{Plasmid}$= 1000 µg*µL / 233.4 µg = 4.3 µL	$V_{Plasmid}$ 4.3 µL

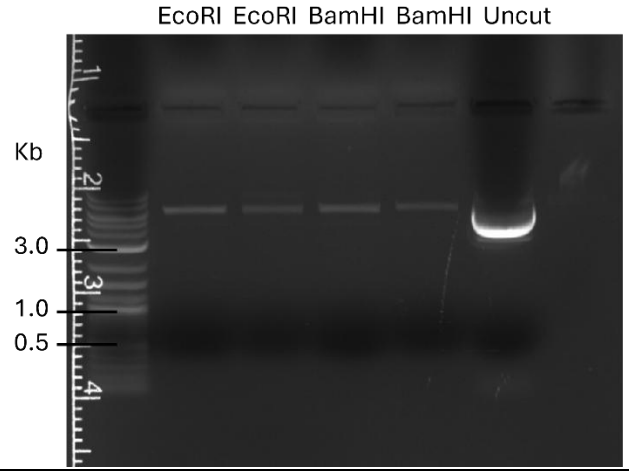
V_{EcoRI}= 2 µL	V_{BamHI}= 2 µL
V_{Buffer}= 5 µL	V_{Buffer}= 5 µL
V_{H2O}= 38.7 µL	V_{H2O}= 38.7 µL

- Heat for 30 min at 37 °C then add 5 µL buffer and 2.5 µL A. Phosphatase added and heated again for 30 min at 37 °C
- Heated to 80 °C

AGE:

- 20 µL from samples 1, EcoRI 2, EcoRI 1, BamHI 2, BamHI Uncut2 + 5 µL loading dye
- Vortex
- 25 µL loaded in AGE:

Markers | 1, EcoRI | 2, EcoRI | 1, BamHI | 2, BamHI | Uncut2



Date	13.03.2025
Who	Noemi, Janina, Yazan
What	Continuation of cultivation: Sample every 1.5 h Proceed: Take ~1.5 mL samples Measure OD600 (dilute directly) -> from 10 o'clock onwards dilute directly 1:200 (first 1:100, then 1:2) Centrifuge the remainder Transfer excess to a separate Eppi Freeze pellet & supernatant

8:30 a.m.	t5	WT	1:100	0.3712		
		pPBEx2	1:100	0.4133		
		pPBEx2+	1:200	0.2590		
10:10 a.m.	t6	WT	1:200	0.4480		
		pPBEx2	1:200	0.2943		
		pPBEx2+	1:200	0.2839		
12:15 p.m.	t7	WT	1:200	0.2317	0.1918	0.1740
		pPBEx2	1:200	0.1884		

			pPBEx2+	1:200	0.2344		
	1:50 p.m.	t8	WT	1:200	0.2282		
			pPBEx2	1:200	0.1880		
			pPBEx2+	1:200	0.1911		
	3:25 p.m.	t9	WT	1:200	0.1768		
			pPBEx2	1:200	0.1751		
			pPBEx2+	1:200	0.2231		

Date	17.03.2025
Who	Noemi
What	Transform Katrin's pK19 into E.coli 100 µL competent cells + 4 µL plasmid 20 min on ice 1 min 42 °C 1 min on ice +800 µL LB medium Shake for 1 h at 37 °C Plating on LB Kan50 -> supernatant + sediment Incubate overnight at 37 °C

Date	18.03.2025
Who	Noemi
What	Precultures for plasmid isolation + cryos 3 x 10 mL LB medium + 10 µL Kan50 Inoculation (approx. 12:15 h) with colony from transformation of 17.03.2025 (*E. coli* DH5α λpir pK19 (Katrin))

Date	19.03.2025
Who	Noemi
What	OD measurement VK 1: 0.3471 CV 2: 0.3147 -> too little for plasmid isolation

Date	20.03.2025
Who	Noemi
What	Precultures 3 x 10 mL pK19 (Katrin*) E. coli* grown on LB with 10 µL Kan50 Plate from 17.03.2015 repainted on grid

Date	21.03.2025
Who	Yazan, Noemi, Julius
ΔmWhat	OD measurement 1: 2.7 2: 2.9 3: 2.7 Cryos VK 1 uses: 750 µL cell suspension laid Mixed with 250 µL glycerol (87%, sterile) Frozen at -70 °C Plasmid isolation ~5.3 mL each pelleted: centrifuged dry Eluted with 30 µL AE, 2 x NanoDrop measurement:

	c[ng/µL]	*A260/280*[ ]	*A260/230*[ ]
1	749.9	2.256	1.947
2	712.3	2.246	1.925
1	693.2	2.245	1.941

Restriction digestion

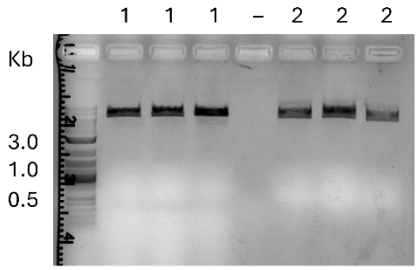
1	EcoRI
	$V_{Plasmid}$= 1000 µg*µL / 749 µg1.3 µL -> 1.3 µL * 2 = 2.6 µL
	V_{EcoRI}= 2 µL
	V_{Buffer}= 5 µL
	V_{H2O}= 38.4 µL
2	BamHI
	$V_{Plasmid}$= 1000 µg*µL / 749 µg1.3 µL -> 1.3 µL * 2 = 2.6 µL
	V_{BamHI}= 2 µL
	V_{Buffer}= 5 µL
	V_{H2O}= 38.4 µL
3	EcoRI + BamHI
	$V_{Plasmid}$= 2.6 µL
	V_{BamHI}= 2 µL
	V_{EcoRI}= 2 µL
	V_{Buffer}= 5 µL
	V_{H2O}= 38.4 µL

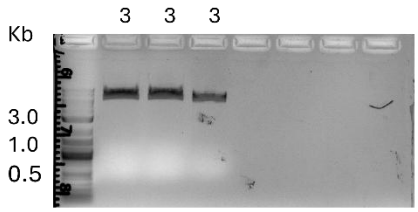
Incubated for 60 min at 37 °C, with phosphatase added after 30 min, followed by heat inactivation at 80 °C for 10 min.

Agarose gel poured: 40 mL 1% agarose + 2 µL PeqGreen

19 µL sample + 4 µL color each

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Ladder	1	1	1	free	2	2	2
Ladder	3	3	3				





Gel clean-up

Empty weight [g]	1a	1.101	2a	1.094	3a	1.096
	1b	1.090	2b	1.097	3b	1.094
	1c	1.101	2c	1.097	3c	1.091
Full weight [g]	1a	1.214	2a	1.213	3a	1.256
	1b	1.212	2b	1.223	3b	1.257
	1c	1,217	2c	1.192	3c	1.259
	1a	0.113	2a	0.119	3a	0.160
	1b	0.122	2b	0.126	3b	0.163
	1c	0.116	2c	0.095	3c	0.168
µL NTI	1a	226	2a	238	3a	320
	1b	244	2b	252	3b	326
	1c	232	2c	190	3c	336

Eluted with 30 µL buffer
NanoDrop measurement:

	c[ng/µL]	*A260/230*[ ]	*A260/280*[ ]
1	28.730	0.8691	2.021
2	23.265	1.319	2.121
1	25.371	1.754	2.056

Date	24.03.25

Who	Alex, Noemi
Was	Preparation of pre-cultures for the rest of the week: Prepare new CG12 stock: 1 ml CaCl2 x 2H2O stock 1 ml MgSO4 stock 1 ml biotin stock --> +920 ml CG12 stock + 80 ml 50% glucose Prepare a total of 20 ml 18.4 µl CG12 stock + 1.6 ml glucose + 20 µl CaCl2 / MgSO4 / biotin + 10 µl Kan 50 +20 µl SEC/ PCA Cultures used 10 µl WT + 10 µl pPEX2 Time: 3:53 p.m. Preculture: CgpPBEx2_oahA & Cg WT auf 10ml CGXII Medium Incubation by 30°C Inoculated at 3:50 p.m. PCR of Katrin's cut pk19: Q5 PCR

Reaction buffer	10µl
10mM dNTPs	1µl
10µM forward Primer	2,5µl
10µM reversed Primer	2,5µl
Template DNA	2µl
Q5 Polymerase	0.5µl
H2O	31,5µl

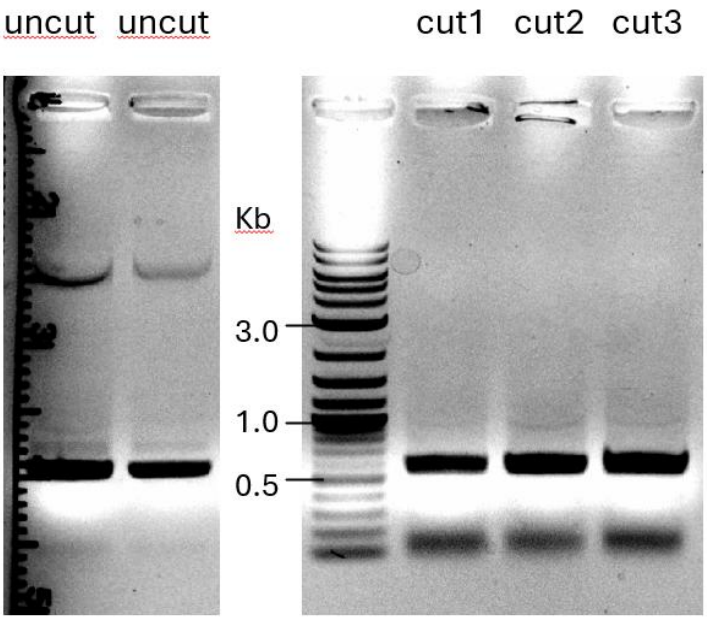
4 batches:

pK19 uncut2 0,5µl DNA
pK19 cut 1 2µl DNA
pK19 cut 2 2µl DNA
pK19 cut 3 2µl DNA

Primer: SecP1, SecP2
T=58°C, delta T=-0.1°C pro cycle

Date	25.03.25

Who	Julius, Lea
Was	AGE 40 ml 1% agarose + 2 µl PeqGreen -> small sled scheme: uncut|uncut|1kb+ ladder|cut1|cut2|cut3 20 µl sample + 5 µl loading dye each



Gibson Assembly for pK19mobsac_oahA_tac
4-6 fragment assembly

	1	2	3
Linear vector (Katrin 2); 29,3 ng/µl	4µl	5 µl	6 µl
Fragment 1 (1:2)	1 µl	1 µl	1 µl
Fragment 2	1 µl	1 µl	1 µl
Fragment 3	1 µl	1 µl	1 µl
Assembly Mastermix	10 µl	10 µl	10 µl
H2O	3 µl	2 µl	1 µl

Transformation in E.coli

- SOP see 19.02.25
- 6 preparations:
    - 100µl competent cells + 10µl Assembly Product 1 to 3
    - 90µl competent cells + 10µl Assembly Product 1&2
    - 80µl competent cells + 10µl Assembly Product 3
- Incubation: 1,5h
- centrifuged, pellet resuspended and plated, incubated at 37°C

Preculture: 2x 25ml cultures (CgWT & Cg pPBEx2_oahA): pPEx2 + 25µl Kan50

0ml Mastermix:

CGxIIstock	46ml
Glucose solution	4ml
PCA	50µl

SEL	50µl

- inoculated with 400µl at 4.45pm

OD of the preculture from 24.03.25: both OD=80

Date	26.03.25
Who	Alex
Was	PCR of the transformed bacteria: Q5 protocol, 50 µl reactions x8

8 batches:

1	Gibson Assembly (GA)1	90+10 colonies
2		
3		
4	GA3	100+10 colonies
5	GA1	90+10 colonies
6		
7	GA2	90+10 colonies
8	pK19uncut (Katrin)	

Mastermix:

Reaction Buffer	80µl
10mM dNTPs	8µl
10µM forward	20µl
10µM reverse	20µl
Q5 Polymerase	4µl

Primer: Seq P1, Seq P2 from Marvin

6.5 µl master mix added to each preparation

3.5 µl H2O added to 1-7

µl DNA + 31.5 µl H2O added to 8

Main culture :6x 50 ml CGXII prepared (see 11.12.24)

- inoculated with 490 µl (WT) or 446 µl (pPBEx2_oahA) at 12:10 p.m. --> OD= 0.5

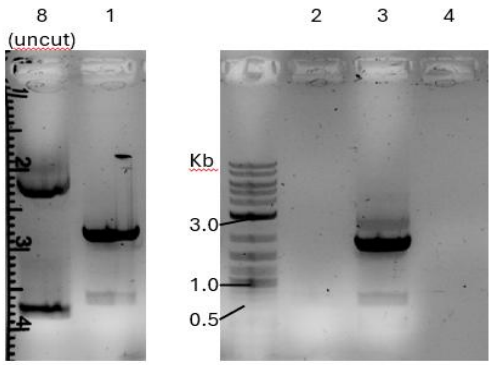
- 2 ml samples taken every 1.5 hours --> 1 ml for OD measurement, 1 ml for freezing
- Cultures:
    - 1. Wild type
    - 2. pPBEx2_oahA
    - 3. pPBEx2_oahA + IPTG by t=0h
    - 4. pPBEx2_oahA + IPTG by t=3h
    - 5. pPBEx2_oahA + IPTG by t=6h
    - 6. pPBEx2_oahA + IPTG by t=24h

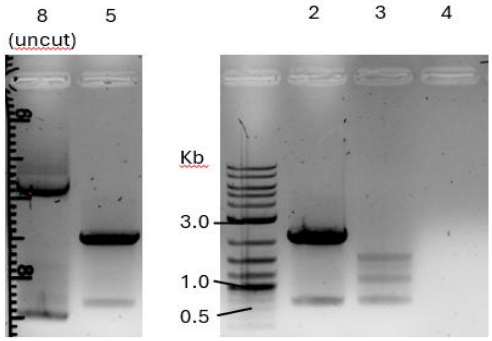
OD 600 Measurment:

Time in h	Dilution	OD_{600}
t=0	-	**1)**0,56 **2)**0,36 **3)**0,45 **4)**0,40 **5)**0,46 **6)**0,48
T=1,5	1:5	**1)** 0,23 **2)**0,31 **3)**0,17 **4)**0,25 **5)**0,17 **6)**0,19
t=3	1:10	**1)**0,08 **2)**0,14 **3**)0,10 **4)**0,14 **5)**0,12 **6)**0,06 **1)**0,2 **2)**0,15 **3**)0,14 **4)**0,14 **5)**0,12 **6)**0,09
t=4,5	1:20	**1)**0,11 **2)**0,12 **3)**0,05 **4)**0,10 **5)**0,06 **6)**0,10
t=6	1:20	**1)**0,27 **2)**0,20 **3)**0,02 **4)**0,05 **5)**0,02 **6**)0,12

Date	27.03.25
Who	
Was	AGE of PCR from 26.03.25: • 20 µl PCR product + 5 µl loading dye each • 1 kB Plus ladder

1	**2**	**3**	**4**	**5**	**6**
8 (uncut)	1	ladder	2	3	4
8 (uncut)	5	ladder	6	7	





Sequencing: Purify fragments (~2800 bp) from gel, send with primers forward/reverse inner seq and outer seq

Gel clean up:

	m without gel	m with gel	m of gel
1	1,0908	1,2797	188,9 mg
5	1,0851	1,2642	179,1 mg
6	1,0890	1,2395	150,5 mg

1	+ 377,8 µl NTI
5	+ 358,2 µl NTI
6	+ 301 µl NTI

- washed twice, with 15 µl H2O eluted

NanoDrop- measurement

	ng/µl	A260/230	A260/280
1	35,391	1,334	1,801
5	37,878	O,543	1,887

6	51,436	0,520	1,865

Sequencing: Fragment concentration should be 10ng/µl, 5µl per tube, 4x in total

1	1: 3,5 diluted	13µl fragment	32,5 µl water	accidentally diluted 1:5
5	1:3,8 diluted	13µl fragment	36,4 µl water	
6	1:5,1 diluted	13µl fragment	53,3 µl water	

Primer: 5µl x3 with c= 5 pmol/ µl (=5 µM)

SecP1	1:2 diluted	10µl Primer + 10µl H2O
SecP2	1:2 diluted	10µl Primer + 10µl H2O
Inner For	1:20 diluted	1 µl Primer + 19µl H2O
Inner Rev	1:20 diluted	1 µl Primer + 19µl H2O

	1	5	6
+SecP1	EF73642755	EF73642742	EF73642738
+SecP2	EF73642756	EF73642741	EF73642737
+inner For	EF73642744	EF73642740	EF73642736
+inner Rev	EF73642743	EF73642739	EF73642735

- all sequencing results negative

Continuation of OD600 measurement from 26.3.25

Time in h	Dilution	OD_{600}
t=21	1:100 1:200 for 3) and 6)	**1)**0,33 **2)**0,40 **3)**0,66 **4)**0,33 **5)**0,44 **6)**0,55 **3)**0,34 **6)**0,28
t=22,5	1:200	**1)** 0,35 **2)**0,36 **3)**0,38 **4)**0,40 **5)**0,37 **6)**0,36
t=24	1:500	**1)**0,12 **2)**0,10 **3**)0,13 **4)**0,13 **5)**0,12 **6)**0,11
t=25,5	1:500	**1)**0,10 **2)**0,10 **3)**0,11 **4)**0,12 **5)**0,18 **6)**0,09
t=27	1:500	**1)**0,07 **2)**0,09 **3)**0,11 **4)**0,12 **5)**0,11 **6**)0,10
t=28	1:500	**1)**0,11 **2)**0,11 **3)**0,12 **4)**0,10 **5)**0,13 **6)**0,07

t=30	1:500	**1)**0,09 **2**)0,01 **3**)0,09 **4**)0,18 **5**)0,07 **6)**0,08
t=48	1:500	**1)**0,14 **2)**0,16 **3**)0,15 **4**)0,17 **5)**0,21 **6**)0,14
Blank diluted	-	**1)**0,15 **2)**0,16 **3)**0,15 **4)**0,17 **5**)0,20 **6)**0,17

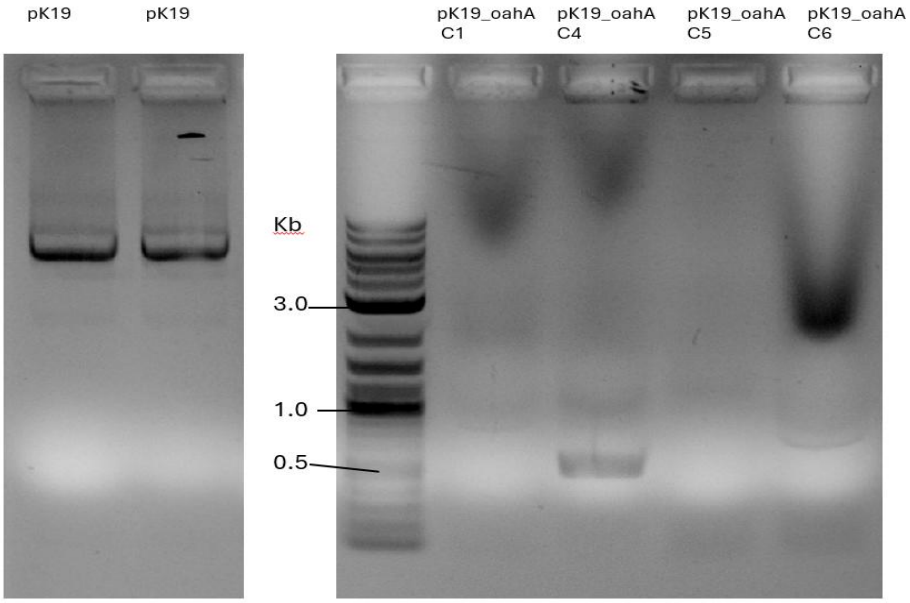
Date	02.04.2025
Who	Julius
What	Over night culture from: GA1 90+10 (1) → Colony from 26.3.25 GA3 100+10 (4) GA1 90+10 (6) GA1 90+10 (5) → 10 ml NB medium → 10 µL Kanamycein Q5 PCR of fragment 1 (upstream homologous) of pK19_oahA_tac (gibson assembly product from 25.3.2025) to check, whether it is preset → comparison to pK19 from Katrin 5 approaches: • 1: pK19 • 2: pK19_oahA_tac from colony + 1 (plate from 26.03.2025) • 3: pK19_oahA_tac from colony + 4 • pK19_oahA_tac from colony + 5 • pK19_oahA_tac from colony + 6 Mastermix for 5 x 50 µL: • Reaction buffer: 50 µL • 10mM dNTPs: 5 µL • 10µM Forward primer: 12,5 µL • 10µM Reverse primer: 12,5 µL • Q5 polymerase: 2,5 µL → 16,5 µL of the mastermix in each tube → To approach 1: 2 µL DNA (pK19 from 21.03.2025) + 31,5 µL ddH_2O → To approach 2-5: one colony + 1,5 µL ddH_2O → Q5 protocol ; primer annealing at 54°C; 1 min elongation

Date	03.04.2025
Who	Amasia, Lea
What	Gelelectrophoresis of the PCR products: • 40 ml 1% agarose + 2 µL PeqGreen • 50 µL PCR product + 10 µL Purpleloading Dye

1	2	3	4	5	6	7

pK19	pK19	Purple 1KbPlus Ladder	pK19_oahA Colony 1	pK19_oahA Colony 4	pK19_oahA Colony 5	pK19_oahA Colony 6

→ 20-25 µL each
→ 30 min running time



→ band at 500 bp of colony 4
PCR Colony 4 for sequencing
batches:

GA3 100+10 colony 4
GA3 100+10 colony 4
GA1 90+10 colony 6
pk19 uncut

Master mix:

- Reaction buffer 40 µL
- 10 µM dNTPs 4 µL
- 10 µM forward 10 µL
- 10 µM reverse 10 µL
- Q 5 Polymerase 2 µL
- in every batch 16.5 µL MM

to 1-3 33.5 µL H2O
to 4 2 µL DNA + 31.5 µL H2O
(Primers: SeqP1, seqP2 by Marvin)

OD of the overnight culture 4: ~ 1.24
cryos of colony 4: 250 µL sterile glycerol + 750 µL cell suspension (6x)

Date	07.04.2025
Who	Noemi

What	AGE of PCR-batches from 3.4.25 Gelextraction Empty weight Eppi: 1,0948 g Full: 1,2877 g Difference: 192,9 mg 386 µl NTI-buffer 2 x cleaning Elution with 15 µL H_2O NanoDrop-Measurement C = 85,157 ng/µL A 260/230 = 2,435 A 260/280 = 1,987 Sequencing PCR-Product should have 10 ng/µL 13 µL DNA + 97,7 µL ddH_2O Primer should have 5 µM SeqP1: 1:2 → 6 µL primer + 6 µL ddH_2O SeqP2: 1:2 → 6 µL primer + 6 µL ddH_2O Inner For: 1:20 → 1 µL Primer + 19 µL H_2O Inner Rev: 1:20 → 1 µL Primer + 19 µL H_2O 5 µL primer-dilution + 5 µL DNA-dilution

K4 + SeqP1	**EF73642734**
+ SeqP2	EF73642730
+ Inner For	EF73642732
+ Inner Rev	EF73642729

Date	14.04.2025
Who	Noemi
What	Preculture for Plasmidisolation 10 mL LB medium in 100 mL flask 10 µL Kan50 Inoculate with 200 µL E.Coli pk19oahA from colony ~16:30

Date	15.04.2025
Who	Noemi
What	OD from preculture (14.04): 11 → ~ 1 ml for plasmidisolation Plasmidisolation: 3 batches, pipetting 1 ml cellsuspension in each 1.5 ml tube Concentration: • 506,431 µg/µL • 542,119 µg/µL • 569,688 µg/µL Q5-PCR: • Comparison of oahA fragment with homologous flanks of colony 4 – SeqP1, SeqP2, Primer

- Negative control with pk19 from Marvin
- 3 batches (1x pk19, 2x pk19 oahA)

Reaction buffer 10 µL	10 µL
10mM dNTPs 1 µL	1 µL
10µM For	2,5 µL
10 µM Rev	2,5 µL
Q5 Pol	0,5 µL
ddH_2O	31,5 µL
DNA	2 µL

0 = pk19

1= pk19 oahA

2 = pk19 oahA

Q5-PCR:

- Synthesizing new fragments

1. Upstream homolog (540 bp) → Primer: Fr.1_FOR, Fr.1_REV
2. oahA_tac (1229 bp) → Primer: Fr.2_FOR, Fr.2_REV
3. Downstream homolog (540 bp) → Primer: Fr.3_FOR, Fr.3_REV

- For 1 and 3 picking C. glut WT Colony: 1 min Elongation; 54°C Annealing

Reaction:

0,5 buffer	10 µL
10 mM dNTPs	2 µL
100 µM FOR	0,5 µL
100 µM Rev	0,5 µL
Q5 Pol.	1 µL
H_2O	33 µL
DMSO	3 µL

→ undiluted primer used

For 2:

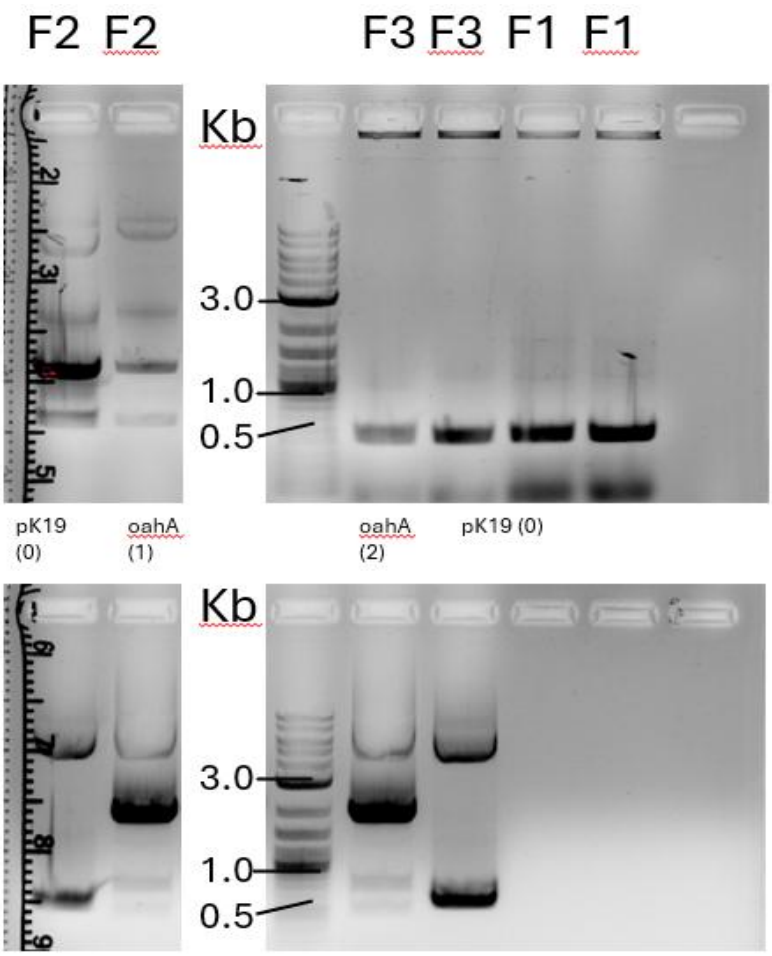
0,5 buffer	10 µL
10 mM dNTPs	1 µL
100 µM FOR	2,5 µL
100 µM Rev	2,5 µL
Q5 Pol.	0,5 µL
H_2O	31,5 µL
DNA	2 µL

Date	16.04.2025
Who	Noemi, Janina
What	AGE: • 40 ml 1% agarose -> 2 µL PeqGreen

- Verifying of fragments synthesized on 15.04
- oahA fragments from 15.04 (from Plasmid from colony 4)

Scheme:

1	2	3	4	5	6	7	8
F2	F2	ladder	F3	F3	F1	F1	
pK19(0)	oahA (1)		oahA (2)	Pk19(0)	tubes broken		



Weighing of tubes	Weighing of tubes with bands	difference
F2 P1: 1,1011g	1,2330	0,1319 → 2635 µL
F2 P2: 1,0964 g	1,2125	0,1161 → 232,2 µL
F3 P1: 1,0950 g	1,2269	0,1319 → 262,8
F3P2: 1,1010 g	1,2329	0,1319 → 263,8
F1 P1: 1,1007 g	1,2549	0,1542 → 308,4
F1P2: 1,1078 g	1,2645	0,1567 → 313,4

Purifying DNA: 15 µL NE used

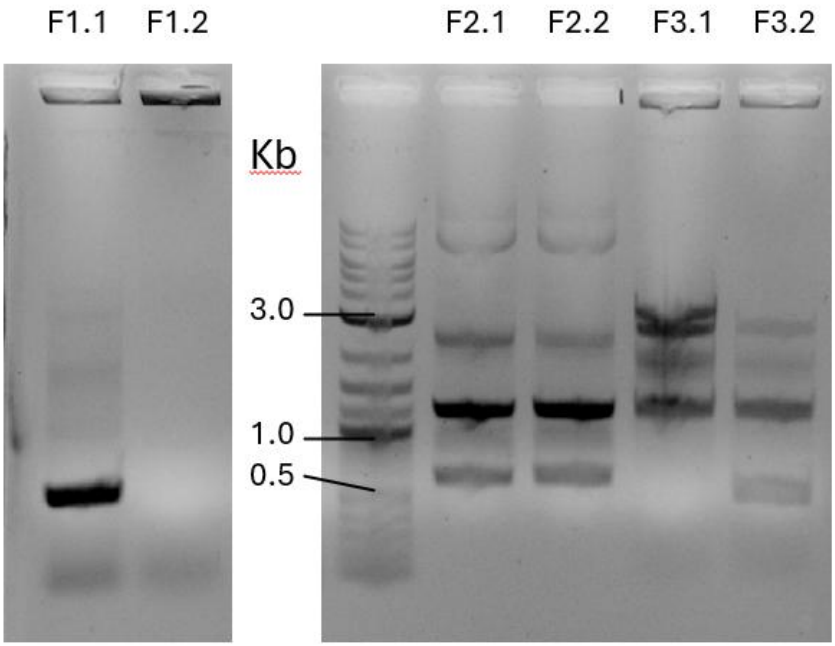
- concentration between 5 ng/µL and 7 ng/ µL → repeat

new Q5 PCR (15.04.2025, 2x every batch)

new AGE

Scheme:

1	2	3	4	5	6	7
F1.1	F1.2	ladder	F2.1	F2.2	F3.1	F3.2



Mass of tubes	Mass of tubes with bands	difference
F1.1 1,0899 g	1,2259 g	136 mg
F2.1 1,0917 g	1,1805 g	88,8 mg
F2.2 1,0945 g	1,1879 g	93,4 mg

Purifying DNA: 15 µL NE used

1: 41,239 µg/µL

2: 40,826 ng/µL

Gibson Assembly
4–6 fragment assembly

	1	2	3
Lin. vector	4 µL	4 µL	6 µL
F1	1.5 µL	1.3 µL	1 µL (diluted 1:2)
F2	3 µL	3 µL	1 µL
F3	1.2 µL	1 µL	1 µL

Mastermix	10 µL	10 µL	10 µL
Water	0.5 µL	0.7 µL	1 µL

Batch 1 &2 with F1 and F2 frpm 17.04 and F3 from previous

Batch 3: F1, F2 and F3 from the same day

Lin vector 1pk19 (Katrin) cut 28,73 ng/µL

Heat shock: 100 µL competent cells + 10 µL Gibson Assembly Product

→ 3 batches

3 plates (spin down)

- Gibson batch 1
- Gibson batch 2
- Gibson batch 3

Component	1	2	3
Lin. vector	4 µL	4 µL	6 µL
F1	1,5 µL	1,3 µL	1 µL
F2	3 µL	3 µL	1 µL
F3	1,2 µL	1 µL	1 µL
MM	10 µL	10 µL	10 µL
H_2O	0,5 µL	0,7 µL	1 µL

Date	24.04.2025
Who	Amelie, Lea
What	Q5-PCR: synthesizing new fragment

- Upstream homolog (540 bp) → Primer: Fr.1_FOR, Fr.1_REV
- oahA_tac (1229 bp) → Primer: Fr.2_FOR, Fr.2_REV
- Downstream homolog (540 bp) → Primer: Fr.3_FOR, Fr.3_REV
- For 1 and 3 picking C. glut WT Colony: 1 min Elongation; 54°C Annealing

Reaction:

0,5 buffer	10 µL
10 mM dNTPs	2 µL
100 µM FOR	0,5 µL
100 µM Rev	0,5 µL
Q5 Pol.	1 µL
H_2O	33 µL
DMSO	3 µL

→ undliuted primer used

- For 2

0,5 buffer	10 µL
10 mM dNTPs	1 µL
100 µM FOR	2,5 µL
100 µM Rev	2,5 µL
Q5 Pol.	0,5 µL
H_2O	31,5 µL
DNA	2 µL

- 1% Agarose
- 5 g Agarose with 500mL TAE-buffern
- Microwave for ca. 6 min. 100°C level 4
- Pour gel with 5 µL pegGREEN

LB Medium:

Bacterio Yeast Extraction	2,5 g/L
Bacterio Tryptone	5 g/L
NaCl	5 g/L
Agar	9 g/L

fill to 500 mL with ddH_2O

→ autoclave

Nanodrop Concentration measurement

(two batches were carry out with 1.2 and 3 each)

2.1: 192,532 ng/µL

2.2: 336,691 ng/µL

1.1: 855,461 ng/µL

1.2: 869,729 ng/µL

3.1: 885,471 ng/µL

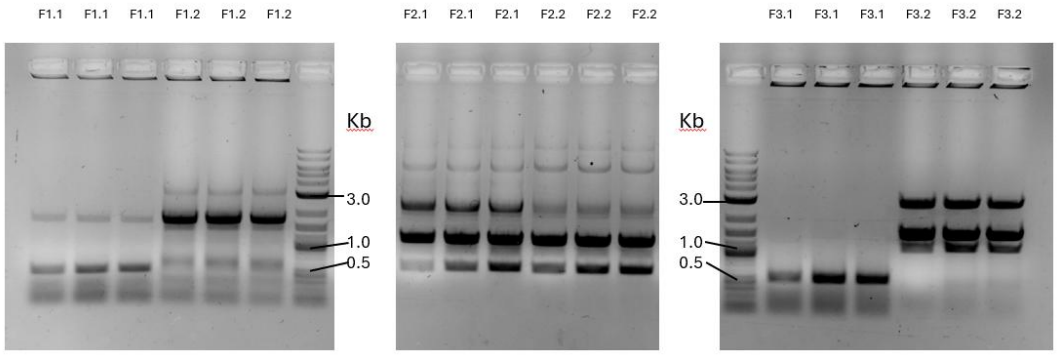
3.2: 993,749 ng/µL

Gelelectrophoresis:

scheme:

1	2	3	4	5	6	7	8	9	10
F1.1	F1.1	F1.1	F1.2	F1.2	F1.2	ladder	F2.1	F2.1	F2.1

11	12	13	14	15	16	17	18	19	20
F2.2	F2.2	F2.2	ladder	F3.1	F3.1	F3.1	F3.2	F3.2	F3.2



Mass of tubes	Mass of tubes with bands	difference	NTI µ
1.1.1: 1025,4 mg	1148,6 mg	123,2 mg	246,4
1.1.2: 1022,9 mg	1124,5 mg	101,6 mg	203,2
1.1.3: 1021,1 mg	1123,0 mg	101,1 mg	202,2
2.1.1: 1026,5 mg	1125,9 mg	95,4 mg	198,8
2.1.2: 1024,0 mg	1097,2 mg	73,2 mg	146,4
2.1.3: 1026,2 mg	1104,0 mg	77,8 mg	155,6
2.2.1: 1026,4 mg	1131,2 mg	104,8 mg	209,6
2.2.2: 1027,5 mg	1124,9 mg	97,4 mg	194,8
2.2.3: 1022,5 mg	1138,3 mg	115,8 mg	231,6
3.1.1: 1020,0 mg	1168,7 mg	148,7 mg	297,4
3.1.2: 1020,0 mg	1201,0 mg	181 mg	362
3.1.3: 1026,5 mg	1206,1 mg	179,6 mg	358,2

Purifying DNA: 15µL NE used

Date	25.04.25
Who	Noemi, Julius
Was	concentrations of the purified fragments:

Sample	Concentration [ng/µL]	$A_{260/230}$	$A_{260/280}$
F1	3,593	0,010	5,106
F2	7,537	0,016	2,893
F3	5,586	0,009	14,840

Pouring gel:

- 500mL LB-agar
- 500µL canamycin

preparing 20 plates

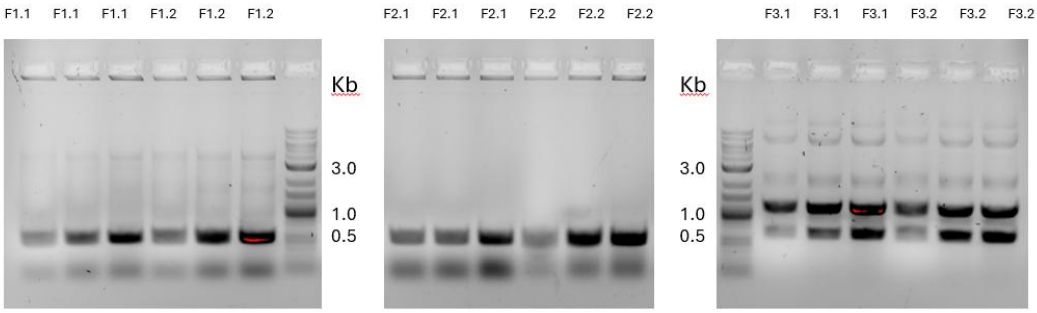
Q5-PCR:

- preparing new fragments
- like protokoll from 15.04.25
- making two samples of each

Gelelectrophoresis like on 24.04.25
scheme: 3x 1.1 3x 1.2 ladder 3x 3.1 3x 3.2 ladder 3x 2.1 3x 2.2

Tubes	Empty weight [mg]	Full weight [mg]	Δm [mg]	NTI [µL]
1-1,2	1091,8	1474,3	382,5	765
1-3,4	1096,1	1454,5	358,4	716,8
1-5,6	1092,4	1463,5	371,1	742,1
2-1,2	1101,4	1317,3	215,9	431,8
2-3,4	1094,5	1303,2	208,7	417,4
2-5,6	1101,8	1393,3	291,5	583
3-1,2	1086,3	1511,6	425,3	850,6
3-3,4	1096,1	1536,4	440,3	880,6
3-5,6	1096,1	1577,0	480,9	961,8

Date	28.04.2025
Who	Noemi. Lea, Paul
Was	Q5-PCR: • preparing new fragments • like protokoll from 15.04.25 • making two samples of each Gelelectrophoresis like on 24.04.25 scheme: 3x 1.1 3x 1.2 ladder 3x 3.1 3x 3.2 ladder 3x 2.1 3x 2.2



Tubes	Empty weight [mg]	Full weight [mg]	Δm [mg]	NTI [µL]
1-1,2	1091,8	1474,3	382,5	765

1-3,4	1096,1	1454,5	358,4	716,8
1-5,6	1092,4	1463,5	371,1	742,1
2-1,2	1101,4	1317,3	215,9	431,8
2-3,4	1094,5	1303,2	208,7	417,4
2-5,6	1101,8	1393,3	291,5	583
3-1,2	1086,3	1511,6	425,3	850,6
3-3,4	1096,1	1536,4	440,3	880,6
3-5,6	1096,1	1577,0	480,9	961,8

Date	29.04.2025
Who	Amelie, Paul
Was	Purify DNA: 15µL NE used Concentrations of purified fragments:

Fragments	Concentration [ng/mL]	$A_{260/230}$	$A_{260/280}$
F1 1-4	5,024	0,015	2,963
F1 5-6	45,579	0,151	1,683
F2 1-4	21,567	0,058	1,756
F2 5-6	18,848	0,07	1,779
F3 1-4	30,842	0,121	1,746
F3 5-6	2,333	0,009	3,07

Dephosphorylation using Antarctic Phosphatase
F1 5-6: ~0,79 µg → ~3,74 µL + 13,26 µL Water
F2 1-4 + 5-6: ~0,4 µg → ~19,79 µL + 0 µL Water
F3 1-4: ~0,17 µg → ~5,51 µL + 11,49 µL Water

Gibson assembly:
using 4-6 fragments

	1	2
Lin. Vector (25,371 ng/µL)	2µL	3µL
Fragment 1	2,5µL	2,5µL

	Fragment 2	2µL	2µL
	Fragment 3	2,5µL	2,5µL
	Assembly MM	10µL	10µL
	Water	1µL	0µL

Transformation in *E. coli* like performed on 19.02.25 (p. 34)
two implementations per approach

Date	30.04.25
Who	Lea, Julius
Was	~0,46M OPD - ~2M HCl solution prepared → There was only 32% HCl → approx. 9 mL used instead of 8

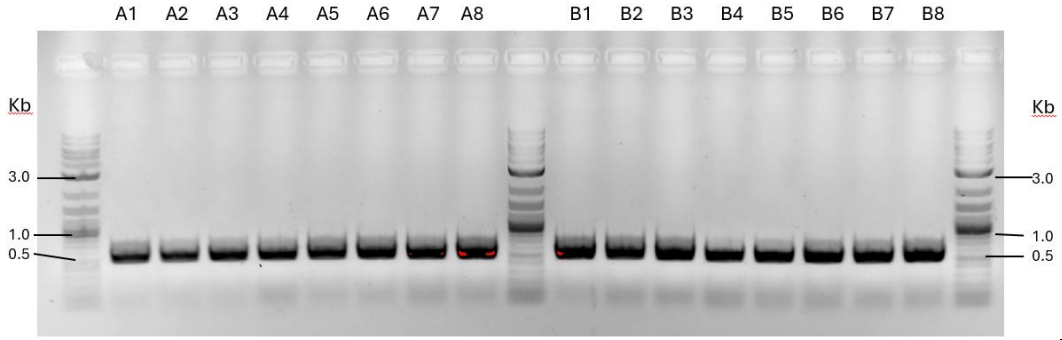
Date	02.05.25
Who	Julius, Lea
Was	Q5 PCR → testing grown colonies from 29.04.25 → from plates GA 1.1 and GA 2.2 8 colonies were taken from each, plated and amplified using colony PCR Preparing mastermix for 16 colonies (17x Q5 protocol, 50µL approach)

Ingredient	Volume [µL]
Reaction buffer	170
10mM dNTPs	17
P1 Seq	42,5
P2 Seq	42,5
Q5 Polymerase	8,5
ddH_2O	570
DNA sample	Inoculated <1

Labeling:
- GA 1.1 → A
- GA 2.2 → B

samples 1 – 8 each

PCR-programm:
1. 98°C 2min
2. 98°C 10sec
3. 58°C 30sec
4. 72°C 01:26min

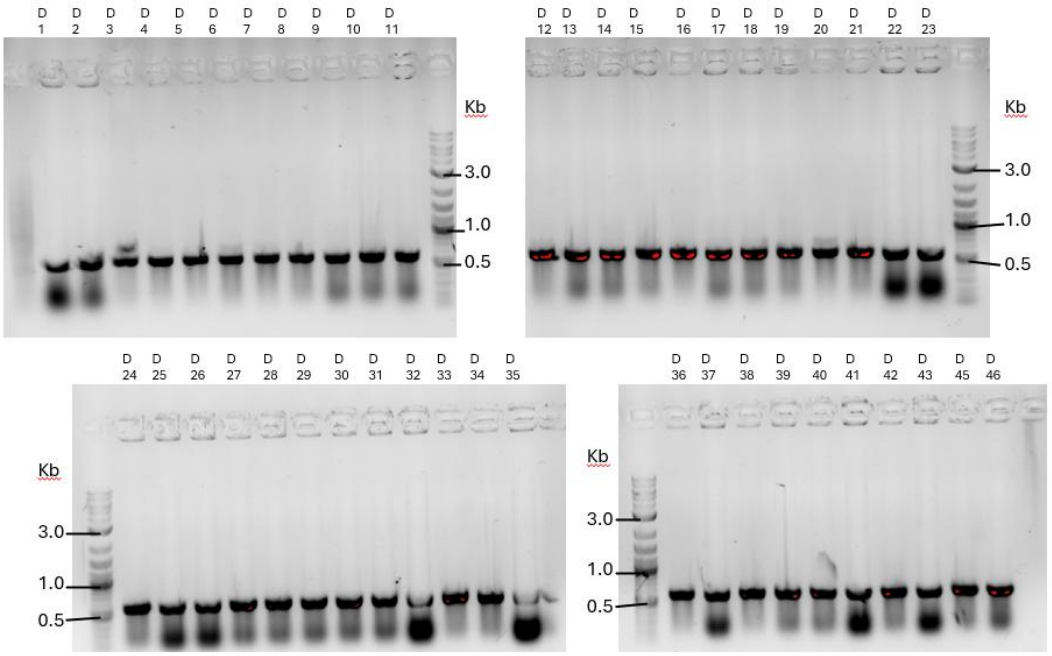
	5. 72°C 2min 6. 4°C ∞ ∞ GE: 100ml Agar sl. + 10µL green • taking 8 25µL samples from A and B with 5µL loading dye 6x • placing 25µL on the gel 

Date	05.05.25
Who	Julius, Amelie
Was	LB medium 1L • Bacto Yeast Extract → 5g • Bacto Tryptone → 10g • NaCl → 10g • Agar → 18g fill ddH_2O to 1L → autoklave

Date	06.05.25
Who	Amelie, Lea, Paul, Julius
Was	Colony PCR using OneTaq polymerase Mastermix (1200µL): • forward primer (Seq P1) → 24µL • reverse primer (Seq P2) → 24µL • One Taq quick load 2x Master Mix with standard buffer → 600µL • H_2O → 552µL Primer sequences: • Seq P1: TAA TGC AGC TGG CAC GAC • Seq P2: CCA TGT AAG CCC ACT GCA → E.coli DH5α λpir pk19-oahA, 29.04.25 plate D, 46 colonies → preparing two grid plates with D1 – D46 colonies → PCR One Taq PCR settings: 1. 94°C 5min 2. 94°C 30sec 3. 52°C 60s 4. 68°C 2:48min 5. 68°C 5min 6. 4°C AGE of the colony PCR-product • 100µL 1% agarose + 5µL PeqGreen • 12µL PCR-product+ 12µL purple 1KB+ ladder in the pockets

Pocket →	1	2-12	13	14-25	26
Lane 1 →	Marker !wrong only to 1KB	PCR-product D1-D11	Marker 1KB+	PCR-product D12-D23	Marker 1KB+
Lane 2 →	Marker !wrong only to 1KB	D24-D35	1KB+ one empty pocket	D36-D43 + D45-D46	

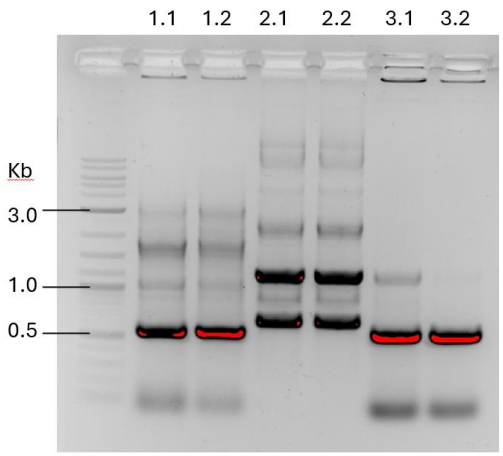
30min, 90V



Date	08.05.25
Who	Julius, Noemi
Was	200mM stock Oxalic acid (50mL): M = 126,07 g/mol (dihydrate) m = 1,2607 g per 50mL 10M NaOH (50mL) 20g filled to 50mL with ddH_2O preparing different oxalic acid standards (10mL each): • 40µM • 60µM • 80µM • 160000µM filling a HPLC glas vial with standard: • 200µL of the 160mM standard + 800µL ddH_2O

	• with M (oxalic acid) = 90,04 g/mol → ~2,88 g/L in vial (ideally between 0,2 and 2 g/L max. 10 g/L)

Date	13.05.25
Who	Lea, Amelie
Was	New Q5-PCR → performed like on 15.04.25 Colony PCR with plate A and B (21 colonies) Mastermix for 600µL: • 10µM forward primer (Seq P1): 12µL • 10µM reverse primer (Seq P2): 12µL • One Taq Quick load 2x MM with standard buffer: 300µL • ddH_2O: 276µL → preparing gridplate with colonies A1-A4 and B1-B17 Gelelectrophoresis with PCR fragments: ladder 1.1 1.2 2.1 2.2 3.1 3.2 1KB+



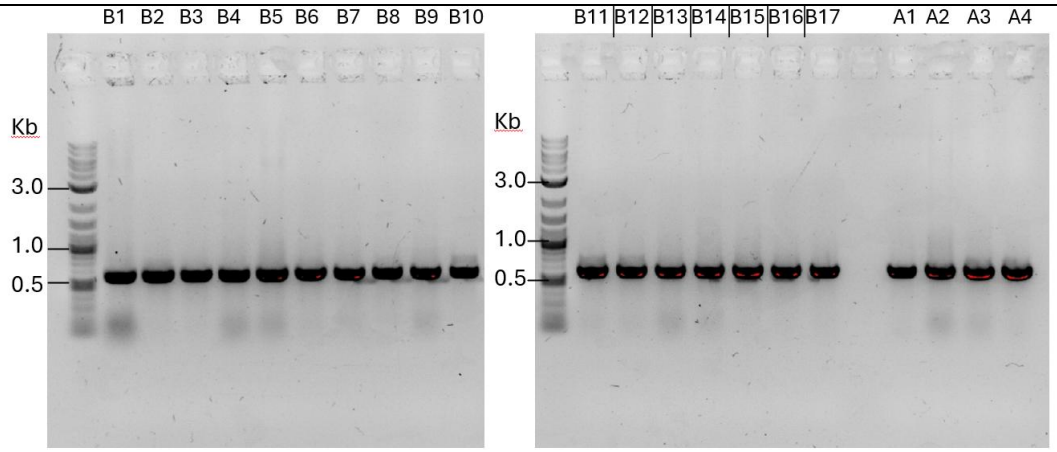
1% agar solution prepared

pruifying the fragments using the protokoll kit
(sample 40µL) for sample 3.2 less then 40µL

AGE colony-PCR-product like p.58

1	2-11	12	13-19	20-23
Purple ladder 1KB+	B1-B10	Purple ladder 1KB+	B11-B17	A1-A4

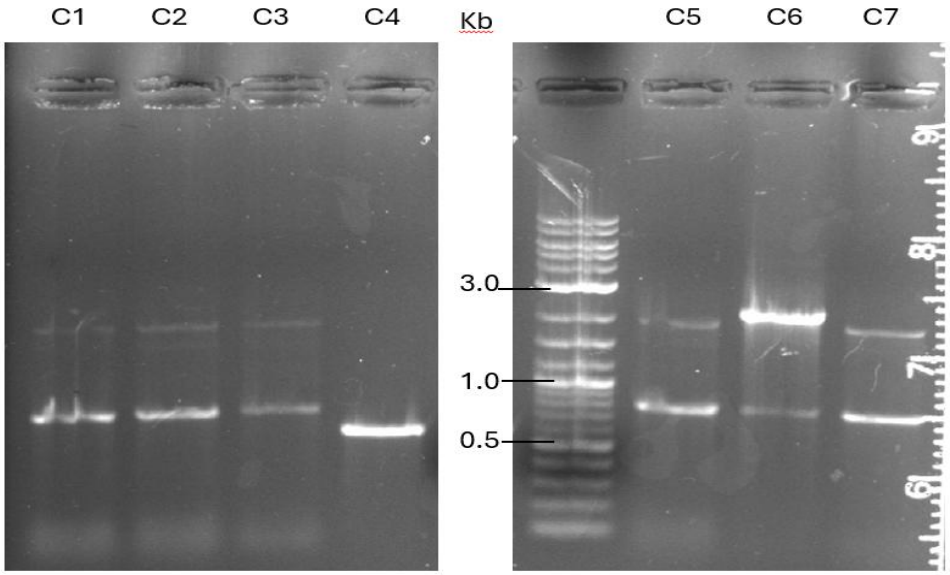
12µL in each pocket
30min, 90V



concentrations of the purified fragments:

Sample	Concentration / ng/µl	$A_{260/230}$	$A_{260/280}$
1.1	211,914	0,864	1,910
1.2	200,944	0,98	1,889
2.1	185,363	1,347	1,9
2.2	141,985	1,21	1,903
3.1	140,652	0,979	1,913
3.2	105,593	0,904	1,927

Date	14.05.2025
Who	Noemi, Lea
What	Gibson-Assembly: • with pK19 (cut) from Katrin 1 and 2 • with new fragments from 13.05.2025 2 Batches: 1): • 10 µl Mastermix • 7 µl pK19 (29.265 ng/µl) • 1 µl fragment 1 (211.941 ng/µl) • 1 µl fragment 2 (185.363 ng/µl) • 1 µl fragment 3 (140.652 ng/µl) 2): • 10 µl Mastermix • 7 µl pK19 (28.73 ng/µl) • fragments like in 1) incubate 1h at 50°C in the thermocycler Transformation in *E. Coli:* • 100 µl competent cells + 10 µl Gibson-Batch 1 and 2 • procedure: see 14.02.2025

Date	15.05.2025
Who	Noemi, Julius
What	Colony PCR: • 7 samples • OneTaq Quick Load protocol • 25 µl reactions • Mastermix: 8 * 25 µl = 200 µl ◦ 10 µM forward primer: 4 µl (seq. P1 from Marvin) ◦ 10 µM reverse primer: 4 µl (seq. P2 from Marvin) ◦ OneTaq Mastermix: 100 µl ◦ ddH2O: 92 µl • picked 7 colonies from 14.05.2025, streaked on grid plate, resuspendet in PCR tubes • PCR program: see 06.05.25 • Gel electrophoresis: • 7 samples from Colony PCR • gel: 40 ml 1% Agar + 2 µl PeqGreen • scheme: • 1|2|3|4|1kb+ ladder|5|6|7 

Date	16.05.2025
Who	Paul
What	Restreaked Colony 6 from 15.05.2024

Date	20.05.2025
Who	Amelie, Lea, Julius, Paul
What	Q5 PCR:

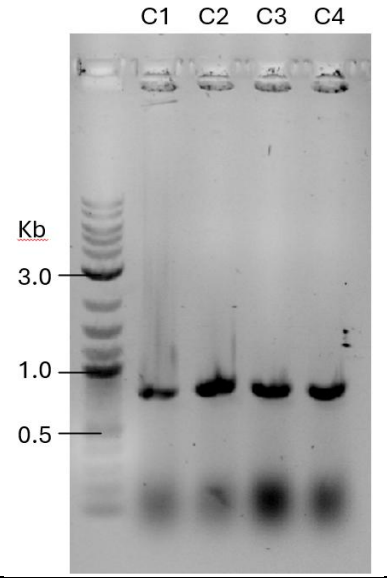
	• 50 µl batch (250 µl) • program: see 02.05.2025 ◦ reaction buffer: 50 µl ◦ dNTPs 10 mM: 5 µl ◦ forward primer Seq, P1: 12.5 µl ◦ reverse primer Seq. P2: 12.5 µl ◦ Q5 polymerase: 2.5 µl ◦ ddH2O: 167.5 µl Transformation of gibson assembly product from 14.05.2025 in *E. Coli* • 100 µl competent cells + 10 µl gibson batch 1 and 2 • program: see 14.02.2025 • incubation at 700 rpm in the thermocylcer Making of oxalate standards: • 3 Standards; 10 ml of each: 2 g/l; 1 g/l; 0.2 g/l

Date	21.05.2025
Who	Amelie, Janina
What	Colony PCR: • 4 colonies • OneTaq Quick Load protocol • 25 µl reaction • Mastermix: 5 * 25 µl = 125 µl ◦ 10 µM forward primer Seq. P1 from Marvin: 2.5 µl ◦ 10 µM reverse primer Seq. P2 from Marvin: 2.5 µl ◦ OneTaq Mastermix: 62.5 µl ◦ ddH2O: 57.5 µl ◦ picked 7 colonies from 14.05.2025, streaked on grid plate, resuspendet in PCR tubes ◦ PCR program: see 06.05.2025 PCR Clean-Up of Q5 product from 20.05.2025: • → see SOP Kit Concentration of fragments:

Sample	Concentration / ng/µl	A260/230	A260/280
1	40.069	0.142	1.874
2	12.650	0.472	3.529
3	36.389	0.452	1.810
4	109.020	1.282	1.823

Sent fragments to sequencing

AGE of Colony PCR product:

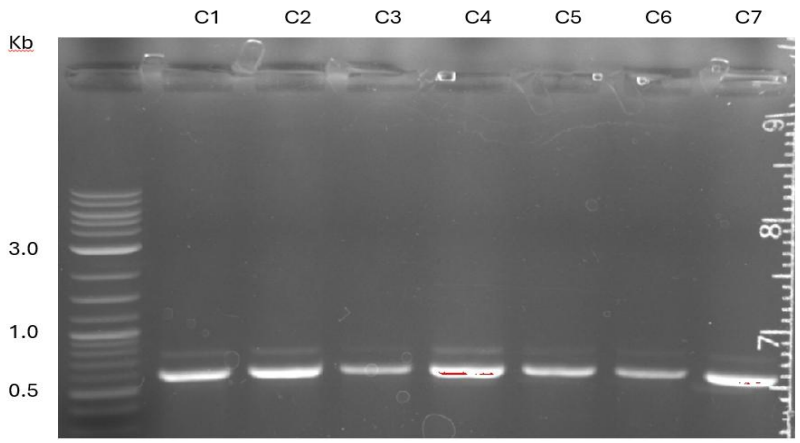
	• 40 ml 1% agarose gel + 2 µl PeqGreen • 12 µl PCR product + 12 µl purple 1kb+ ladder in pockets • scheme: ladder|colony 1|colony 2|colony 3| colony 4| colony 5 • 30 minutes; 90 V 

Date	27.05.2025
Who	Amelie
What	Q5 PCR: • procedure: see 15.04.2025 • new annealing temperature: ◦ fragment 1: 67 °C ◦ fragment 2: 65 °C ◦ fragment 3: 66 °C

Date	28.05.2025
Who	Noemi
What	AGE: • making 1% agarose solution (1 g agarose + 100 ml 1xTAE) • scheme: F1.1|F1.2|1kb+ ladder|F2.1|F2.2|1kb+ ladder|F3.1|F3.2 Gel cleanup: • empty weight: ◦ 1: 1.0997 g ◦ 2: 1.1025 g ◦ 3: 1.0937 g • weight w/ gel: ◦ 1: 1.2685 g → Δm = 0.1688 g ◦ 2: 1.2430 g → Δm = 0.1405 g ◦ 3: 1.2725 g → Δm = 0.1788 g measured concentrations: • upstream(1): 62.024 ng/µl • oahA (2): 48.150 ng/µl • downstream: (3): 74.037 ng/µl

Date	02.06.2025

Who	Noemi
What	Gibson-Assembly: • new fragments + pK19 (cut) * 3 • reaction calculated with NEBuilder Protocol Calculator: ◦ pk19: 7.0 µl ◦ oahA: 0.8 µl ◦ downstream: 0.9 µl (diluted 1:4 → 19 ng/µl) ◦ upstream: 0.5 µl (diluted 1:2 → 31 ng/µl) ◦ ddH2O: 0.8 µl ◦ Assembly MM: 10.0 µl • pK19 not enough: → 2.7 µl (3) (25.371 ng/µl); 3.5 µl (2) (~29 ng/µl); 0.8 µl (1) (~29 ng/µl) → only used tube 1 Transformation in *E. Coli*: • 1* 100 µl competent cells + 10 µl assembly product • 1* 100 µl competent cells + 7 µl assembly product

Date	03.06.2025
Who	Paul, Amelie, Julius
What	Colony PCR: • 7 colonies • OneTaq Quick Load protocol • 25 µl reactions • Mastermix: ◦ 10 µM forward primer: 4 µl ◦ 10 µM reverse primer: 4 µl ◦ OneTaq Mastermix: 100 µl ◦ ddH2O: 92 µl • picked 7 colonies from 14.05.2025, streaked on grid plate, resuspendet in PCR tubes • PCR program: see 06.05.2025 AGE of PCR products: • 10 ml 1% agarose + 2 µl PeqGreen • scheme: 1kb+ ladder|1|2|3|4|5|6|7 

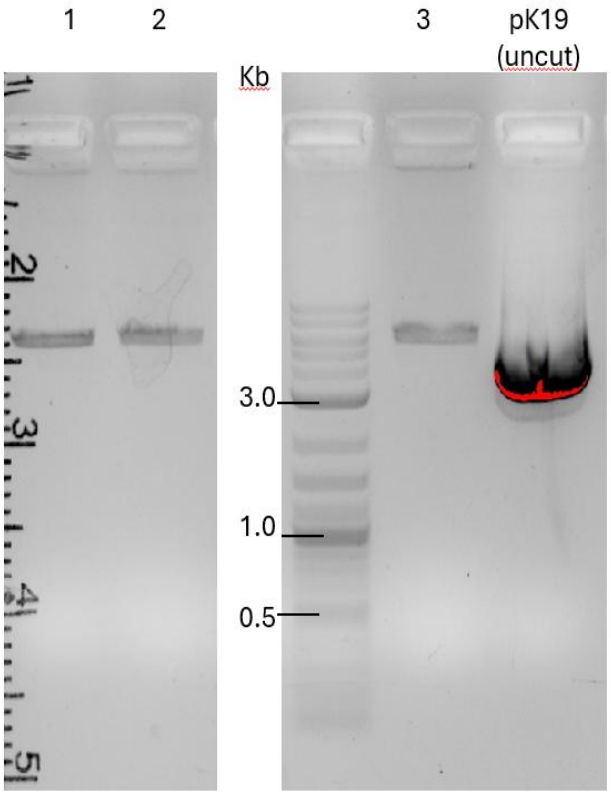
Date	10.06.2025
Who	Julius
What	Making M9 media: • 100 ml (1M) CaCl2•2 H2O: 14.7 g → filter sterilized • 100 ml (1M) MgSO4•7 H2O: 24.65 g → filter sterilized • 500 ml 5x M9 salt solution: ◦ 21.28 g Na2HPO4•2H2O ◦ 7.5 g KH2PO4 ◦ 1.25 g NaCl ◦ 2.5 g NH4Cl ◦ add 500 ml ◦ →autoclave Restriction digestion: • plasmid concentrations: ◦ 1: 749 ng/µl ◦ 2: 712 ng/µl ◦ 3: 693 ng/µl • Mix for restriction digestion: ◦ 2000 ng plasmid (1: 2.67 µl + 48.3 µl ddH2O; 2: 2.81 µl + 48.2 µ ddH2O; 3: 2.89 µl + 48.1 µl ddH2O) ◦ 2 µl BamHI ◦ 2 µl EcoRI ◦ 5 µl 10x buffer ◦ add H2O 50 µl • 30 minutes at 37 °C, 2 µl phosphatase + 6 µl phosphatase buffer each • 30 minutes at 37 °C, then ~5 minutes at 0 °C → freezer Preculture: • 8 ml ddH2O (sterile) • 2 ml 5x M9 • 80 µl glucose (50 g/ 100 ml) • 20 µl 1M MgSO4 • 10 µl kanamycin • 1 µl 1M CaCl2 • 50 µl *E. Coli* • →140 rpm at 37 °C

Date	11.06.2025
Who	Lea, Noemi
What	Making LB-medium: 4x50 mL ddH_2O 40 ml 5xM9 10 ml Glc (50%) 400 µl 1M $MgSO_4$ 100 µl Kanamycin 50 µl $CaCl_2$ 5 µl

	→ in 2 of the 4: 100 µl IPTG (0,269 g/ml) • Preculture from 10.06.2025 has not grown → two new precultures • 10 ml each of the 50 ml LB medium without IPTG 4 new Precultures: • 2 with M9 medium: 10 ml M9 medium + 100 µl E.coli pEKEx2_oahA K9/ pPBEx2_oahA K4 • 2 with LB medium: 10 ml LB medium + 10 µl Kan50 + 100 µl µl E.coli pEKEx2_oahA K9/ pPBEx2_oahA K4 → Inoculated at 5:00 pm & incubation at 37°C, 140 rpm

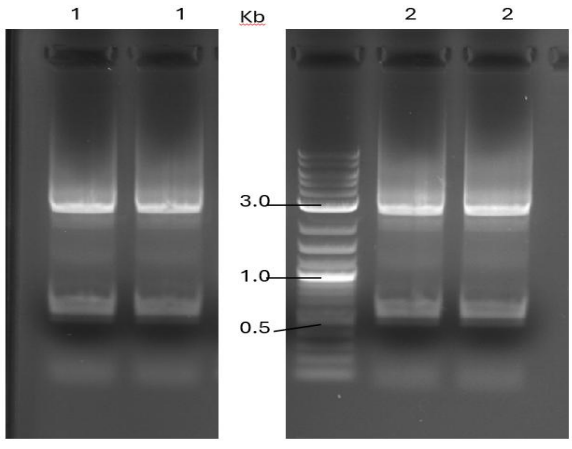
Date	12.06.2025
Who	Noemi
What	• Measurement of the OD values of the preculture → all too low • streaking out E.coli cryos (pEKEx2_oahA K9 and pPBEx2_oahA K4) on LB-Kan50 plates

Date	13.06.2025
Who	Julius, Noemi
What	Gibson Assembly: <table><tr><td>pK19 (48 ng/ml)</td><td>36 µl</td></tr><tr><td>oahA</td><td>0,8 µl</td></tr><tr><td>Upstream 1:2</td><td>0,5 µl</td></tr><tr><td>Downstream 1:4</td><td>0,9 µl</td></tr><tr><td>ddH_2O</td><td>4,2 µl</td></tr><tr><td>Mastermix</td><td>10 µl</td></tr></table> Gel electrophoresis with linearized pK19 from 10.06.2025: <table><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>1)</td><td>2)</td><td>Ladder</td><td>3)</td><td>PK19 uncut</td></tr></table>



Date	14.06.2025
Who	Noemi
What	Q5-PCR: • 2x 50 µl with Gibson Assembly products from 13.06.2025 • procedure: see 19.02.2025 Gel electrophoresis with PCR product:

1	2	3	4	5
1)	1)	Ladder	2)	2)



Date	16.06.2025
Who	Noemi
What	Electroporation: • 125 µl competent C. Glut WT from Marvin + 29,6 µl Gibson Assembly product (from 13.06.2025) → transfer in cuvette • start electroporation • 2. culture see SOP • incubation at 30°C, 130 rpm

Date	17.06.2025
Who	Amelie
What	Making BHI + sucrose plates: • 20% sucrose solution: dissolve 200 g sucrose in 500 ml ddH_2O → fill up to 1 l • BHI solution: dissolve 74 g BHI in 1 l ddH_2O + 36 g agar → autoclave both solutions → 78 BHI + sucrose plates poured Centrifuge cellculture for 2 min at 11.000 g → discard supernatant except 100 µl → plate on Kan50 plates

Date	23.06.2025
Who	Noemi, Amelie
What	• 6x 10 ml CGXII medium in 100 ml shaking flask + 10 µl Kan50 • inoculated at 17:30 with 100 µl C.glut pPBEx2_oahA / pEKEx2_oahA • incuabtion at 30°C, 130 rpm

Date	24.06.2025
Who	Amelie
What	Preculture II: • 6 x 25 ml CGXII medium in 250 ml shaking flasks with: • 23 ml CGXII stock solution • 2 ml 50% glucose solution • 25 µl SEL • 25 µl PCA • 25 µl Kan502 Adding 500 µl Pre I from 23.06.2025 at 14:45 → discard Pre I

Date	25.06.2025

Who	Amasia,Lea
What	Inoculate main culture: OD_{600} = 0,5

	Dilution	OD_{600} with dilution	OD_{600} without dilution	Volume for inoculation
1	1:200	0,20	40	0,625 µl
2	1:200	0,19	38	0,658 µl
3	1:200	0,20	40	0,625 µl
4	1:200	0,19	38	0,658 µl
5	1:200	0,20	40	0,625 µl
6	1:200	0,20	40	0,625 µl

→ Dilution in two steps: first 1:10 with 900 µl ddH_2O, second 1:20 with 950 µl ddH_2O
→ 1-3: pEKEx2_oahA
→ 4-6: pPBEx2_oahA

OD measurement of the main culture:

		OD600					
Zeit	Zeit [h]	pPBEx2_oahA #1	pPBEx2_oahA #2	pPBEx2_oahA #3	pEKEx2_oahA #1	pEKEx2_oahA #2	pEKEx2_oahA #3
11:45	0	0.4	0.45	0.42	0.44	0.42	0.42
13:15	1.5	0.9	0.85	0.95	0.85	0.95	1.05
15:00	3.25	1.6	1.5	1.6	1.7	1.6	1.6
16:15	4.5	2	2.2	2.6	2.8	2.2	2.4
17:45	6	3.4	4	4.6	4.6	3.2	3.8
19:15	7.5	6	7.6	8.6	8	6	7.2
09:45	22	54	52	56	54	72	68
11:15	23.5	46	50	44	54	64	58

→ Sample volume: 1,5 ml (except at t=1,5: 2 ml from sample number 4 and 5)

Induction with IPTG (0,268 g/ml = 1,125 mM) at t=3:
→ 15 min to late

Culture	Final concentration	Inoculation

		in mM	volume in µl
	1 (46,5 ml)	2	82,7
	2 (46,5 ml)	2	82,7
	3 (46,5 ml)	2	82,7
	4 (46,0 ml)	2	81,8
	5 (46,0 ml)	2	81,8
	6 (46,5 ml)	2	82,7

→ Centrifugation of the samples at 11000 g for 2 min
→Freeze supernatant at -21°C

Preculture for plasmid isolation:

- 10 ml LB medium in 100 ml shaking flask + 10 µl Kan50
- Inoculate with 200 µl E.coli pK19 at 18:30; incuabtion at 37°C

Date	26.06.2025
Who	Amelie, Lea
What	OD_{600} measurement at 9:45 and 11:45 (see 25.06.2025) → Centrifugation of the samples at 11000 g for 2 min →Freeze supernatant at -21°C Plasmid Easy Pore Kit (NucleoSpin): → OD_{600} = 4,6 → 3 ml used for the kit → freeze at -21°C

Date	27.06.2025
Who	Lea, Paul, Noemi
What	Restriction digestion of pK19 (see 26.06.2025): • 2000 ng plamsid → c(pK19) = 280,4 ng/ µL → V = 7,1 µL • 2 µL HindIII • 2 µL PstI • 5 µL 10x buffer • 33,9 µL ddH_2O → incubate 30 min at 37°C → add 2 µL phosphatase + 6 µL phosphatase buffer → incubate 30 min at 37°C → heat inactiavtion at 80°C for 2 min CLean up with PCR/gel clean up kit: • all restriction digestions on one column • eluation with 30 µL

	• Nano drop measurement: 115,2 ng/µL

Date	30.06.2025
Who	Lea
What	• 2x 500 ml BHI + agar: 18,5g BHI + 9g Agar each • 2x 500 ml 50% glucose solution (550 g/l) • 2x 500 ml Saline: 1g NaCl + 1l ddH_2O Pouring plates: • 20 BHI Kan7,5 (75 µL Kan50 in 500 ml medium) • 20 BHI Kan25 (250 µL Kan50 in 500 ml medium)

Date	01.07.2025
Who	Dani, Noemi
What	Making solutions for phosphate precipitation (→ HPLC) → see SOP, 100 ml each

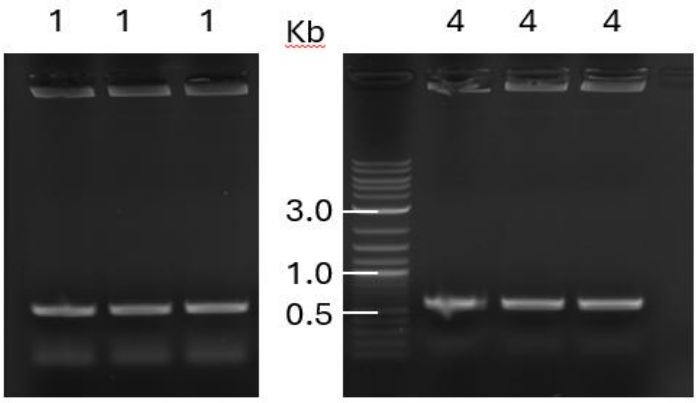
Date	02.07.2025
Who	Noemi
What	• 100 g/l Glc - stock: 100 ml, 10,88 g (with HPLC water) • 10 g/l oxalic - acid stock: 100 ml, 1,4 g • 4 g/l oxalic acid - stock: 4 ml 10 g/l stock + 6 ml ddH_2O • 4 g/l oxalic acid - stock + 4 g/l glc - stock: 4 ml 10 g/l oxalic acid- stock + 400 µL 100 g/l glc - stock + 5,6 ml ddH_2O • Set 2l flow agent (see HPLC script) • Dilute samples from the cultivation from KW26 1:50 → 20 µL sample + 50 µL 4 g/l oxalic acid – stock + 930 µL ddH_2O (final concentration: 0,2 g/l) Perepare standards:

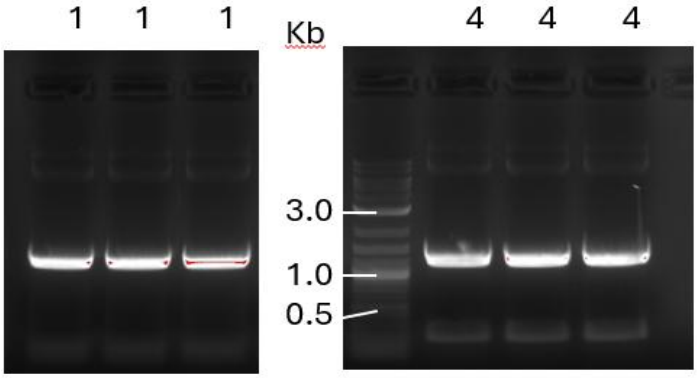
mg/l	4 g/l + 4 g/l stock in µL	HPLC water in µL
2000	500	500
1000	250	750
400	100	900
200	50	950
100	25	975
40	10	990
20	5	995
0	0	1000

Date	**02.07.2025**
Who	**Lea, Amasia**
What	Primer vorbereitet: angegebene Menge an H2O hinzugegeben 1:10 auf 10 µM verdünnt: 450 µL H2O + 50 µL Primer PCR für neue Fragmente: Upstream = 1 Primer SB17 + SB18, WT als Template LacI = 2 Primer SB19 + SB20, pPKEx7.oahA aus K9 als Template oahA = 3 Primer SB21 + SB22, pPKEx2.oahA Downstream = 4 Primer SB23 + SB24, WT ○ Jeweils 50 µL Reaktion angesetzt wie im Skript beschrieben ○ PCR Einstellungen S.14 Fragment 1: 72 °C 50 sec Fragment 2: 68 °C 50 sec Fragment 3: 66 °C 50 sec Fragment 4: 72 °C 50 sec Glc-Lösung sterilfiltriert AGE: 40 mL 1% Agarose + 2 µL PeqGreen Schema:

1	2	3	4	5	6	7
1	1	1	ladder	4	4	4
-	-	-	-	-	-	-
2	2	2	ladder	3	3	3

- Je 21 µL Proben geladen dafür
- 50 µL PCR Ansatz mit 12,5 µL loading buffer vermischt
- Banden sahen gut aus





Gel clean-up:

1 – 255 mg -> 510 µL NTI
2 – 243 mg -> 486 µL NTI
3 – 250 mg -> 500 µL NTI
4 – 295 mg -> 570 µL NTI

- 2x gewaschen und eluiert mit 30 µL Puffer

Date	**03.07.2025**
Who	**Lea**
What	

	ID	Name	Länge	[c]
Fragmente	F1	Upstream	551 bp	42,5 ng/µL
	F2	LacI	1320 bp	81,6 ng/µL
	F3	oahA	1269 bp	89,2 ng/µL
	F4	Downstream	561 bp	66,3 ng/µL
Vektor	Pk19		5722 bp	115,2 ng/µL

Gibson Assembly: NE Builder Protokoll

Component	Volume [µL]	Volume x3 [µL]

Pk19	1,5	4,5
F1	0,4	1,2
F2	0,5	1,5
F3	0,4	0,9
F4	0,3	1,2
ddH2O	6,9	20,7
NE Builder MM	10	30

- 3 PCR Tubes: 60 min 50 °C (Marvins Gibson Programm Thermocycler)

Transformation in E.coli: (Protokoll vom 10.02.2025, S. 34)

Statt 1 µL Plasmid 8 µL
700 rpm

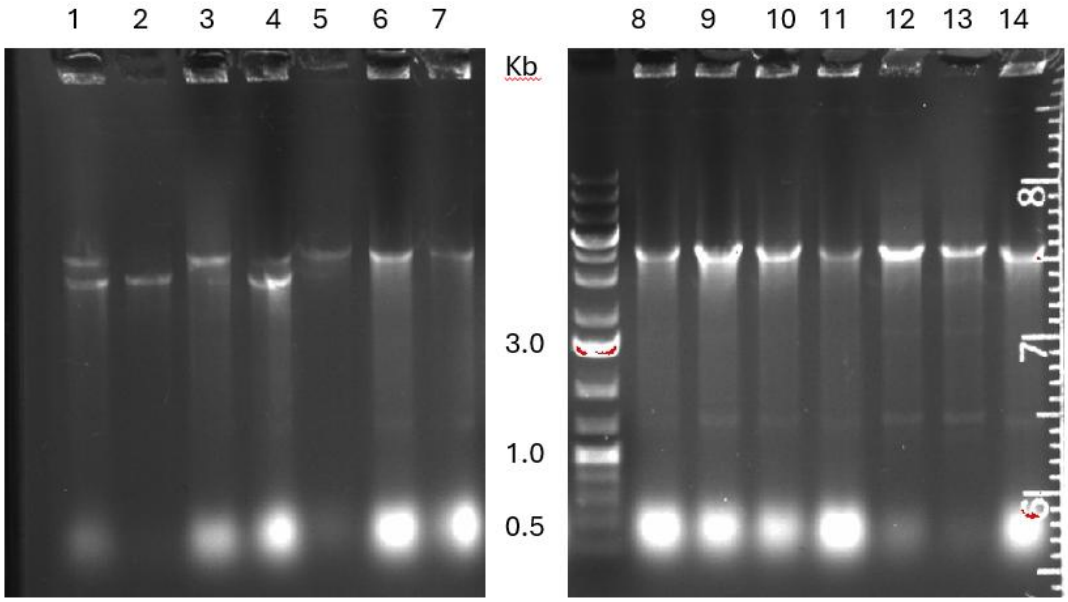
PCR des Gibson Assembly Produkts: Q5 PCR (S.34)

Reaction Buffer	10 µL	Step:	Temp:	Time:
10 mM dNTPs	1 µL	1	98 °C	30 sec
Sec P1	2,5 µL	2	98 °C	10 sec
Sec P2	2,5 µL	3	58 °C	30 sec
Template DNA	8 µL	4	72 °C	50 sec *zu kurze: nochmal mit 2 min 6: für 4178 bp
Q5	0,5 µL	5	72 °C	120 sec
Nuclease free water	29,5 µL	6	4 °C	unendlich

Date	**04.07.2025**
Who	**Amelie**
What	PCR Colony PCR Forward Primer 7,5 µL Reverse Primer 7,5 µL

	MM 187,5 µL Water 375 µL

Date	**07.07.2025**
Who	**Amelie**
What	Colony Gelelektrophorese 1 % Agarose + 2 µL PeqGreen samples 1-14 ~10 µL 1 2 3 4 5 6 7 ladder 8 9 10 11 12 13 14 90 V, 30 min -> instead of 4170 bp just ~2800 bp



NEBuilder DNA Assembly: NE Builder Protokoll

Component	[ng/µL]	Volume [µL]
pK19 PstI from Marvin	43,79	4
F1	42,5	1,5
F2	81,6	1,5
F3	89,2	1,5
F4	66,3	1,5
ddH2O	43,79	4
NE Builder MM		10

	2 PCR Tubes 60 min (Marvins Gibson programm Thermocycler)

Date	**09.07.2025**
Who	**Amelie**
What	Made 0,9 % Saline 1 L Made 5 M KOH solution 500 mL

Date	**14.07.2025**
Who	**Julius, Amelie**
What	CGXII Stock (NH4)2SO4 20g/L x 920 mL=18,4 g Urea 5g/L x 920 mL=4,6 g KH2PO4 1g/L x 920 mL=0,92 g K2HPO4 20g/L x 920 mL=0,92 g MOPS 42g/L x 920 mL=38,64 g • Fill to 700 mL with ddH2O • Set pH7 with 5M KOH • Fill to 920 mL with ddH2O • autoclave

Date	**16.07.2025**
Who	**Lea, Amelie**
What	Building the DASGIP reactors

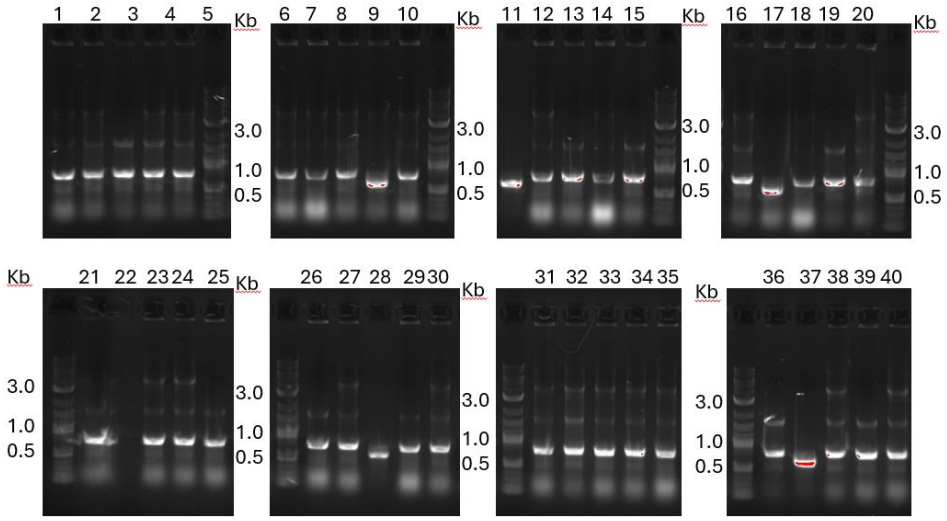
Date	**18.07.2025**
Who	**Noemi, Lea**
What	Colony PCR with Marvin‘s transformed E.coli: pK19.oahA (Plate: 5µL, ÜS from 17.07.2025) 40x 12,5 µL reactions on PCR well plate colonies striked on grid and incubated at 37 °C Well plate scheme:

	1	2	3	4	5
A	1	2	3	4	5
B	6	7	8	9	10
C	11	12	13	14	15
D	16	17	16	19	20
E	21	22	23	24	25
F	26	17	26	29	30
G	31	32	33	34	35
H	36	37	36	39	40

AGE:

100 mL 1 % Agarose + 5 µL PeqGreen
10 µL in 30er combs
Loading scheme:

1 2 3 4 5 ladder 6 7 8 10 ladder
ladder 21 22 23 24 25 ladder …



keine positiven Klone

Date	**20.07.2025**
Who	**Noemi**
What	~10 PM: 5 mL 2TY-medium inoculated with WT and pKBEx2.oahA (C. glut) (50 µL Cryo each)

Date	**21.07.2025**
Who	**Daniel, Ayleen, Amelie, Lea, Noemi**
What	• 9 AM: Preculture-2 on 50 mL CGXII-Medium (inoculated with complete Pre-1) • 6 DASGIP-reaktors assembled • pH-probes calibrated • reactors autoclaved with 700/662 mL CGXII-stock • Cleaning-in-place with 2x water and 1x medium (acid&base) • reactors connected to DASWARE control system • DO-probes calibrated • Offgas analytics started Medium for Pre-3 prepared: • 4x 300 mL CGXII-Medium in 3 L baffled flasks • 2x with, 2x w/o Kan50 OD Pre-3: ~3 • Too low: should be min 15 after 12 h growth • Experiment delayed by one day • new Pre-1 inoculated, this time with 200 µL Cryo • offgas analysis switched off

Date	**22.07.2025**
Who	**Noemi**
What	• 9 AM: Pre-2 inoculated with 5 mL Pre-1 each • ~09:30 PM: Pre-3 inoculated on OD=2 ○ Filled needed cell-suspension volume of Pre-2 in falcons and centrifuged at 4960 g for 2 min ○ Discarded supernatant ○ Pellet resuspneded in 5 mL NaCl (0,5x) ○ Centrifuged again ○ Discarded supernatant ○ Pellet resuspended in Pre-3 Medium (CG12) and Pre-3 inoculated ○ → Cells washed, to avoid carry over of 2TY-medium in reactors • Offgas analysis restartet

Date	**23.07.2025**
Who	**Julius, Noemi, Daniel, Ayleen, Amasia, Paul, Theresa**

What

- Sterile supplements for each reactor filled in influx-bottles:

R1 and R2	R3 and R4	R5	R6
1 mL MgSO4	1 mL MgSO4	1 mL MgSO4	1 mL MgSO4
1 mL Biotin	1 mL Biotin	1 mL Biotin	1 mL Biotin
1 mL CaCl2	1 mL CaCl2	1 mL CaCl2	1 mL CaCl2
80 mL Glc	80 mL Glc	80 mL Glc	80 mL Glc
1 mL PCA	1 mL PCA	1 mL PCA	1 mL PCA
1 mL TES	1 mL TES	1 mL TES	1 mL TES
1 mL Kan50	-	1 mL Kan50	-

- OD of Pre-3 measured:
  - pPBEx2.oahA C.g. #1 and #2
  - C.g. WT #1 and #2
  - pPBEx2.oahA #1 and WT #2 used (same OD -> better comparable)
  - needed inoculation volumes filled in inlet bottles and ddH2O added to reach 1 L end volume

R1	42 mL pPBEx.oahA #1	2105 mL H2O
R2	42 mL pPBEx.oahA #1	2105 mL H2O
R3	42 mL WT #2	255 mL H2O
R4	42 mL WT #2	255 mL H2O
R5	42 mL pPBEx.oahA #1	173 mL H2O
R6	42 mL WT #2	173 mL H2O

- reactors inoculated ~10 PM
- T0 sample taken, every hour OD and glucose measured, supernatant-samples stored at -20 °C, after ~3h (01:10 PM) induced with 2 mM IPTG: R1, R2, R6
- Antifoam added manually if needed
- From OD ~ 3 BTM – samples taken
- Cultivation continued until ~ 0:30 AM

Date	24.07.2025
Who	Noemi, Daniel, Theresa
What	DASGIP autoclaved, dismantled, rinsed

Date	28.07.2025
Who	Noemi, Amasia
What	3 oxalate standards (phosphate precipitated) placed in HPLC3 for Johannes Kan50 stocks produced (50 mg/mL) -> sterile filtered, in Eppis, frozen

	BTM tubes from the DASGIP trial weighed

Date	29.07.2025
Who	Noemi, Julius
What	PCA stocks produced (see Marvin's SOP) Potassium oxalar stock, NH4OH stock & CaCl2 stock prepared for new oxalate assay

Date	30.07.25
Personal	Daniel Dorta, Amasia Rocha, Noemi Reusing
Protocol	**Reagent preparation:** 1. 1mM potassium-oxalate ($K_2C_2O_4$) stock-solution. (For calibration standards) [Use of 18,423mg of potassium-oxalate-monohydrate dissolved in 100ml ddH_2O] 2. 100mM calcium-chloride ($CaCl_2$) stock-solution. (For the precipitation of oxalate in the samples) [1,4702g of $CaCl_2$ dissolved in 100ml ddH_2O] 3. 0.05mM ammonium hydroxide (NH_4OH) stock-solution. (For resuspension of the pelleted oxalate) [112µl 25% NH_4OH in a total volume of 30ml ddH_2O] **Protocol Steps for each sample/ calibration standard:** 1. Use of 350µl DasGip sample supernatant (pre-centrifuged to extract the cells). 2. DF 1:2 with 350µl ddH_2O 3. Adding of 140µl $CaCl_2$ stock- solution 4. 15min reaction time at RT for the precipitation of calcium-oxalate (The calcium ions from the $CaCl_2$ should have formed complexes with the oxalate, displayed as a white precipitation in the eppi) No precipitation was seen. 5. Centrifugation at 10.000G/ 5min RT 6. Pellet wash and resuspension in 80% Ethanol (Did not really work, calcium-oxalate complexes are difficult to re-disolve) 7. Centrifugation at 10.000G/ 5min RT 8. Pellet drying at RT/37°C 9. Redissolving in 300µl 0.05mM NH_4OH stock-solution (did not work) 10. Samples should have then be loaded onto a UV-96-well microplate in triplicates (100µl each), but resuspension of the pellet in NH_4OH did not work. 11. Calibration Standard dilutions

Vol. 1mM $K_2C_2O_4$-stock [µl]	Vol. ddH_2O matrix [µl]	Endvol. [µl]
100	900	1000
80	920	1000
65	935	1000
50	950	1000

35	965	1000
25	975	1000
15	985	1000
10	990	1000
5	995	1000
BLK	1000	1000
BLK without precip.	1000	1000

TECAN Spark® Configurations:

- Mode: Absorbance / Endpoint measurement
- Wavelengths: 210, 212, 214, 216 nm (each measured individually or as a series) + reference measurement at 300 nm (if available) to monitor baseline absorption
- Plate type: 96-well, UV-transparent (e.g., polystyrene-free or quartz, Greiner UV-Star)
- Reading direction: From top
- Measurement position: Central (usually set automatically)
- Temperature: RT (22–25 °C)
- Shaking: Linear, duration 3 s, frequency 282 rpm (default), amplitude 1.5 mm (low/medium prevents spilling, timing = before measurement only, plate remains in reader during shaking, no eject required; performed ~3–5 s before measurement)
- Optical path length: 3–3.5 mm at 100 µL volume
- Measurement interval: Not relevant (single measurement per well)
- Blank correction: Optional; measure blank (BLK) in a dedicated well (corrected A = A_measured – A_BLK). This corrects for matrix absorption without oxalate, ensuring specific oxalate absorption.
- Replicates: Triplicates per standard, sample, and control (100 µL per well)

Software Settings (Tecan i-control / Magellan, etc.)

- Plate definition: 96-well UV plate
- Wavelength mode: Multiple λ (210–216 nm)
- Read mode: Absorbance (endpoint)
- Measurement position: Center
- Shake before read: Yes, 3 s, low intensity.
- Measurement time: Auto or 50–100 ms per well
- Filter set: UV (no filter slide required)

Evaluation main errors and key considerations:

- Use more $CaCl_2$ (about 150µl)
- Chill Ethanol for longer periods (increases effiency when removing the matrix and stops re-dissociation

	- A solution for the resuspension of oxalate needs to be found.

Date	10.08.25
Personal	Daniel Dorta, Amasia Rocha
Protocol	Same procedure as in the protocol done on the 30.07.25 only 0.1mM HCL was used for the resuspension of the pelleted oxalate. **Evaluation main errors and key considerations:** Resuspension did not work out! Search for new methods!

Date	11.08.2025
Who	Janina, Amasia, Lea
What	LB medium produced Dried plasmid: 369 ng -> resuspend with 7.38 µL for 50 ng/µL NanoDrop: 2.4 ng/µL

Date	12.08.2025
Who	Lea, Amelie
What	After vortexing again: NanoDrop Plasmid: 4 ng/µL Heat shock in E. coli 60 µL cells and remaining plasmid (3.5 µL) Plated on 2 Amp plates

Date	15.08.2025
Who	Amelie, Amasia
What	Colony PCR with 19 colonies

- Master mix for n = 19

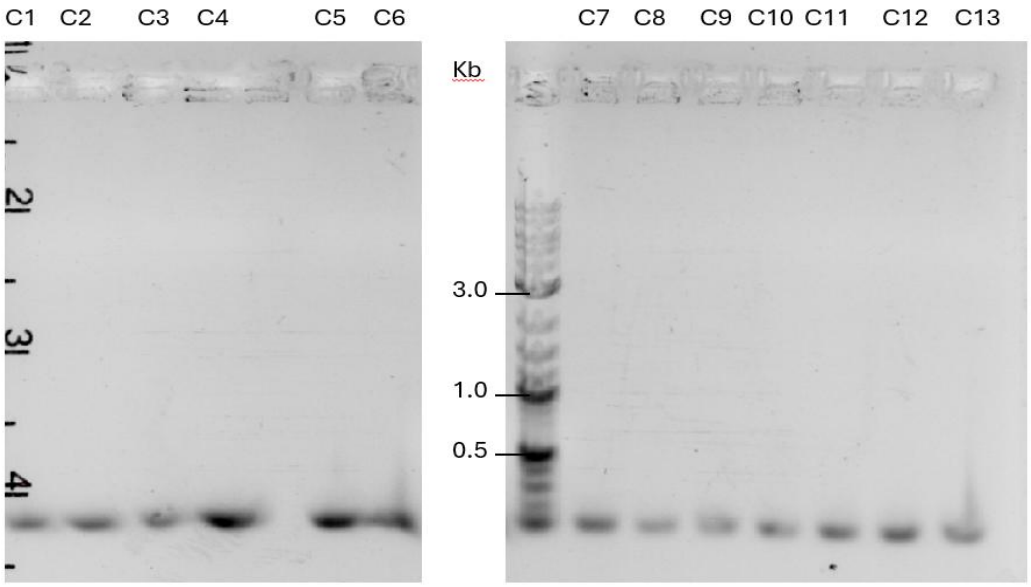
10 µM forward primer (SB17)	10 µL
10 µM reverse primer (SB24)	10 µL
OneTaq Quick Load 2x MM	250 µL
Nuclease-Free Water	230 µL

- Colony PCR program (PCR device iGEM-Lab: my -> COLONY)

Initial Denaturation	94°C	5 minutes	
Denaturation	94°C	30 seconds	30 cycles
Annealing	67°C	60 seconds	
Elongation	68°C	3 min 38 s	
Final Extension	68°C	5 minutes	
Hold	4°C	∞	

Date	18.08.2025
Who	Amasia, Lea
What	AGE • 40 mL 1% agarose + 2 mL Peqgreen • 10 µL of sample and ladder • 30 min, 90 V

1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	2	3	4	5	6	1KB+L adder	7	8	9	10	11	12	13



Date	20.08.2025
Who	Amelie, Amasia
What	• oCPC stock prepared according to protocol (50 ng/mL)

Date	25.08.2025
Who	Amasia, Julius
What	oCPC stock diluted: 0.75 mM (50 mL) • 50 mL Tris buffer: 0.605 g dissolved in 50 mL • 40 mL Tris + 0.478 mL of 50 mg/mL CPC stock • Check pH, should be 10.5, otherwise adjust with HCl or NaOH • Fill to 50 mL with TRIS buffer • check pH again • Experiment: Test with Controls Oxalate Assay

Wells A1 – C10		Wells D1 – F10
CaCl2 [µL]	H2O [µL]	As per protocol potassium oxalate + H
0	1000	
2.5	997.5	
5	995	
7.5	992.5	
10	990	
15	985	

		20	980		
		30	970		
		40	960		
		50	950		
		• 15 min precipitation with 100 µL CaCl2 (100 mM), centrifuged for 10 min, no pellet was visible • 50 µL instead of 30 µL (as in the protocol) CPC			

Date	25.08.25
Personal	Dani, Amasia, Julius
Protocol	**Calcium-Oxalate precipitation combined with oCPC (o Cresolphthalein Complexone)** **Objective:** Oxalate is precipitated through 100mM $CaCl_2$-stock solution, the calcium is added in a 10-30% excess. After the precipitation the excess calcium ions react with the chromogenic dye, Cresolphthalein Complexone, to form a colored complex. This complex absorbs light, and its absorbance is directly proportional to the Ca^{2+} concentration. This allows for an indirect determination of the oxalate concentration. **Reagent preparations:** 1. 0.1 M Tris Buffer (pH 10.5) a. Weigh out 0.6057 g of Tris base. b. Dissolve in approximately 40 mL of deionized water. c. Adjust pH to 10.5 using NaOH. d. Bring volume up to 50 mL with deionized water. 2. 7.5 mM CPC Stock Solution a. Start with a 50 mg/mL CPC stock solution. b. Calculate the volume needed to make 50 mL of a 7.5 mM solution. c. Molar mass of CPC: 636.62 g/mol. d. Mass of CPC needed: 7.5 mmol/L × 0.05 L × 636.62 g/mol ≈ 0.2387 g. e. Volume of 50 mg/mL stock needed: 0.2387 g / (50 mg/mL) = 4.774 mL. f. Pipette 4.78 mL of the 50 mg/mL CPC stock into a 50 mL volumetric flask. g. Add approximately 40 mL of 0.1 M Tris buffer. h. Adjust pH to 10.5 using NaOH. i. Bring volume up to 50 mL with 0.1 M Tris buffer. j. Store as the working CPC stock solution. - 100mM potassium-oxalate ($K_2C_2O_4$) stock-solution. (For calibration standards) - 100mM calcium-chloride ($CaCl_2$) stock-solution. (For the precipitation of oxalate in the samples) [1,4702g of $CaCl_2$ dissolved in 100ml ddH_2O]

Protocol Steps for each sample/ calibration standard:

1. Plate Setup (96-well plate)

- The plate should accommodate standards, controls, and samples.
- Standards: A series of known Ca2+ concentrations for a calibration curve.
- Controls: Reagent blank (no sample), media blank (no Ca2+), and positive control (known Ca2+).
- Samples: Unknown samples to be analyzed.
- Perform all measurements in triplicates for accuracy.

2. Pipetting Scheme

- Standards:
  - Pipette 100 µL of each Ca2+ standard into designated wells.
  - Add CPC working solution.
- Controls:
  - Negative Control: 100 µL of Tris buffer (pH 10.5). Add CPC. This serves as the reagent blank.
  - Positive Control: A known concentration of Ca2+ solution. Add CPC.
- Samples:
  - Add a known volume of sample and CaCl2 solution to the wells.
  - Mix and allow precipitation of calcium oxalate.
  - Transfer 100 µL of the supernatant to a new well.
  - Add CPC working solution.
- Measure the absorbance (Optical Density, OD) at a wavelength appropriate for the CPC complex (e.g., 550-570 nm).
- Subtract the OD of the reagent blank from all sample and standard readings to correct for background.

TECAN Spark® Configurations:

- Mode: Absorbance / Endpoint measurement
- Wavelengths: 210, 212, 214, 216 nm (each measured individually or as a series) + reference measurement at 300 nm (if available) to monitor baseline absorption
- Plate type: 96-well, UV-transparent (e.g., polystyrene-free or quartz, Greiner UV-Star)
- Reading direction: From top
- Measurement position: Central (usually set automatically)
- Temperature: RT (22–25 °C)
- Shaking: Linear, duration 10s, frequency 282 rpm (default), amplitude 1.5 mm (low/medium prevents spilling, timing = before measurement only, plate remains in reader during shaking, no eject required; performed ~3–5 s before measurement)
- Optical path length: 3–3.5 mm at 100 µL volume
- Measurement interval: Not relevant (single measurement per well)
- Blank correction: Optional; measure blank (BLK) in a dedicated well (corrected A = A_measured – A_BLK). This corrects for matrix absorption without oxalate, ensuring specific oxalate absorption.

	- Replicates: Triplicates per standard, sample, and control (100 µL per well) Software Settings (Tecan i-control / Magellan, etc.) - Plate definition: 96-well UV plate - Wavelength mode: Multiple λ (210–216 nm) - Read mode: Absorbance (endpoint) - Measurement position: Center - Shake before read: Yes, 10s, low intensity. - Filter set: UV (no filter slide required) **Evaluation main errors and key considerations:** • CPC excess: Ensure a 10- to 100-fold molar excess of CPC relative to Ca2+ to guarantee the reaction is not limited by the dye. • pH: The reaction requires a stable pH of 10.5. Use a pH meter to confirm and adjust the buffer if necessary. • Precipitation: The formation of Ca-oxalate precipitate can cause a matrix effect. Using the supernatant helps to minimize this interference. • Consistency: Ensure consistent volumes and incubation times across all samples and standards. • 9-Point reads turn out to be counterproductive introducing more data and increasing variability. • Well-A1(100µM) OD= 0,3601 ; Well-A2(0µM) OD= 0,3590. There is minimal difference between the OD´s which leads to the conclusion that the precipitation is inefficient or not working.

Date	27.08.2025
Who	Amasia, Amelie
What	• CaCl2 + H2O and oxalate + H2O again as prepared the day before • Precipitation with 150 µL CaCl2, centrifuged for 15 min, supernatant in extra Eppi and vortexed -> again no pellet visible, maybe CaCl2 and oxalate solution bad? • A1 - C10: CaCl2 + H2O 2nd experiment • D1 - F10: Oxalate + H2O, precipitated supernatant 2nd experiment • G1 - H10: Oxalate + H2O, precipitated supernatant 1st experiment • D11 – H12: CaCl2 + H2O 1. Experiment K4–K5 duplicate • C11 K6 1st attempt C12 K7 1st attempt B11 K8 1st attempt B12 K9 1st attemptA11 K10 1st attempt

Date	27.08.25

Personal	Daniel Dorta Bechel, Amasia Rocha
Protocol	**Objective:** To repeat the calcium-oxalate assay to verify precipitation and determine concentrations. **1. Procedure** • Setup: CaCl2 and oxalate solutions were prepared as on the previous day. • Precipitation: 150 µL of CaCl2 were added to the sample and allowed to stand for 15 minutes. • Centrifugation: The samples were centrifuged for 15 minutes. • Supernatant: The supernatant was transferred to a new Eppendorf tube and weighed. **2. Plate Layout (96-Well Plate)** The plate was arranged as follows: • Wells A1 to C10: CaCl2+H2O (2nd attempt). • Wells D1 to F10: Oxalate + H2O with precipitated supernatant (2nd attempt). • Wells G1 to H10: Oxalate + H2O with precipitated supernatant (1st attempt). • Other Wells: Contained samples from the 1st attempt for repetition. **3. Observations and Notes** • Precipitation: As in the first attempt, **no pellet was visible again**. • Calculation: In a well with 2.5 µL of CaCl2 and 997.5 µL of water, the amount of substance is nCa=2.5×10−8 mol. • CPC Ratio: For the measurement, the amount of CPC should be at least 10 times the maximum amount of calcium in the well, i.e., nCPC≥10×nCa2+.

Date	04.09.25
Personal	Daniel Dorta, Amasia Rocha, Lea
Protocol	**Objective:** To quantify the concentrations of Oxalate in sample the resulting calcium (Ca2+) after a calcium-oxalate precipitation is determined using a colorimetric assay. The protocol is designed to ensure a sufficient molar excess of the indicator for accurate measurement. **Reagent and Sample Preparation:**

- M Tris Buffer (pH 10.5)

    - Weigh 0.6057 g of Tris base.
    - Dissolve in approximately 40 mL of deionized water.
    - Adjust pH to 10.5 using NaOH.
    - Bring volume up to 50 mL with deionized water.
    - Re-verify the pH after final volume adjustment.

    - 0.75 mM CPC Working Solution

        - Start with a 50 mg/mL CPC stock solution.
        - Pipette 0.478 mL of the 50 mg/mL CPC stock into a 50 mL volumetric flask.
        - Add 40 mL of the 0.1 M Tris buffer.
        - Adjust pH to 10.5 using NaOH (a small volume, typically 0.1-1 mL).
        - Bring final volume up to 50 mL with 0.1 M Tris buffer.
        - Store as the working CPC solution.

- 1 mM Calcium Chloride ($CaCl_2$) Stock Solution
- 1 mM Oxalate (Ox) Stock Solution

Plate Setup (96-well plate)

- The plate is organized to accommodate standards, controls, and samples. The assay is performed in triplicates.
- Wells A1 to C10 (Standards):
    - These wells will be used to create a calibration curve.
    - Pipette varying volumes of 1 mM $CaCl_2$ and ddH_2O to a total volume of 1000 µL according to the table below.

$CaCl_2$ (µL)	*ddH_2O (µL)*
0	1000
2.5	997.5
5.0	995.0
7.5	992.5
10.0	990.0
15.0	985.0
20.0	980.0
30.0	970.0
40.0	960.0
50.0	950.0

- Wells D1 to F10 (Test Samples):
    - Pipette varying volumes of Oxalate and water to a total volume of 1000 µL.
    - After the initial reaction and centrifugation (Step 2.1), transfer 100 µL of the supernatant to these wells.

Initial Reaction Setup (in Eppendorf tubes)

- Add 100 µL of 1 mM CaCl2 stock to the tube. This provides a 10% molar excess of Ca2+ for the highest standard.
- Mix thoroughly and allow the calcium oxalate to precipitate for 15 minutes.
- Centrifuge the tube at 10.000G for 10 minutes to pellet the precipitate. (No pellet was noticed)

Transfer and Final Reaction

After precipitation, a portion of the supernatant is transferred to a reaction well for the colorimetric measurement.

1. Carefully transfer 100 µL of the supernatant to a reaction well.
2. Add 50µL of the 0,75 mM CPC stock solution. This ensures a 20-fold molar excess of CPC over the remaining calcium.
3. Mix the contents of the well through 10sec shaking in the TECAN.
4. Measure the absorbance (OD) of the solution at 550-570nm.

Key Molar Ratios and Calculations

The success of this protocol relies on these specific molar ratios.

- Moles of Oxalate added (nox): 1×10^{-7} mol
- Moles of Calcium added (nCa): 1.1×10^{-7} mol
- Molar excess of Calcium: 10%
- Volume of CPC needed: 26.4 µL to ensure a 20x molar excess over the calcium remaining in the supernatant after precipitation.

Evaluation main errors and key considerations:

- Precipitation: No visible pellet was observed after centrifugation. This suggests the initial concentrations might be too low to form a visible precipitate, or the reaction conditions need optimization.

- Assay Volume: The final volume of the solution to which CPC is added was 50 µL, not 30 µL as initially planned. This change should be noted in all future protocols to ensure consistency.

Date	05.09.2025
Who	Amasia, Amelie
What	Experiment with lower calcium concentration and new solutions (store at 4 °C from now on) → 10% excess instead of 100 x → 110 µL of 1 mM CaCl2 for precipitation in Eppi → 26.4 µL of 7.5 mM CPC in well ! Precipitation did not work well -> no pellet visible

	1	2	3	4	5	6	7	8	9	10	11	12
A			Ca2+ after precipitation of oxalate with 26.4 µL CPC 7.5 mM Triplicates									
B												
C												
D			Ca2+ after precipitation of oxalate with 100 µL CPC 7.5 mM									
E			Ca2+ after precipitation of oxalate with 50 µL CPC 7.5 mM									
F			Ca2+ + H2O with 26.4 µL CPC 7.5 mM Triplicates									
G												
H												

- Calibration curve with calcium in water, same volumes as with potassium oxalate (both 1 mM)

Date	05.09.25
Personal	Daniel Dorta, Amasia Rocha
Protocol	**Objective:** To investigate precipitation using a lower calcium concentration and new solutions. **1. Procedure** • Preparation: New, fresh solutions were used. • Assay Setup: o Samples A-E: 110 µL of 1 mM CaCl2 was added to an Eppendorf tube for precipitation. Afterwards, 26.4 µL of the 7.5 mM CPC solution was pipetted into the well. o Samples F-H: Samples were tested with 50 µL of CPC. o Samples A-B: Samples were tested with 400 µL of CPC. o Samples C-E: Samples were tested with 100 µL of CPC. **2. Plate Layout** • All samples were prepared in triplicates to ensure reliability. • The tests included: o Ca2++H2O with 26.4 µL of 7.5 mM CPC.

	○ Ca2+ after precipitation of Oxalate. **3. Observations and Notes** • Precipitation: The precipitation did not work well, and no pellet was visible. • Calibration Curve: A calibration curve was created with calcium in water, using the same amount and volume as with oxalate (both at 1 mM). • Other Tests on the same day: ○ General tests for oxalate precipitation with calcium were performed. ○ Test Setup: 30 µL aliquots of the Calcium Oxalate mixture (from 10 - 400 µL) were topped up to 1000 µL with 100 µM CaCl2 and water. ○ Reaction: 30 minutes of precipitation, followed by 10 minutes of centrifugation at 10,000 g. ○ Result: No pellet was observed in any of these samples. ○ pH Influence: After adding 50 µL of 0.1 M NaOH, a small, normal pellet became visible.

Date	08.09.2025
Who	Dani, Amasia, Noemi
What	General tests for the precipitation of oxalate with calcium: • In 30 µL steps of 10 µL – 400 µL 1 mM potassium oxalate + 100x excess CaCl2 (10 µL – 400 µL 100 mM CaCl2) made up to 1000 µL with H2O • 30 min precipitation, 10 min, 10000 g centrifuged (same for all subsequent samples) • no pellets • Repeat the same procedure after adding 50 µL of 0.1 M NaOH each ○ no pellets • Further tests with potassium oxalate:

1 mM potassium oxalate [µL]	100 mM CaCl2 [µL]	H2O [µL]	
100	300	600	
100	100	800	Old reager from 29.07
100	500	400	
100	500	400	
400	500	100	
3000	3000		

○ No pellets anywhere
• 250 µL sample (t9 2) + 250 µL H2O + 100 µL 100 mM CaCl2
○ Gave a pellet (probably calcium oxalate and cell components there:
• Sample centrifuged (10 min, 10000 g)
○ Gave a pellet (probably cell components)
• Supernatant from the Smaple (250 µL) + 250 µL H2O + 100 µL 100 mM CaCl2
○ Gave a pellet (which should have only been calcium oxalate)
• 10 µL 4 g/L oxalic acid + 890 µL H2O + 100 µL 100 mM CaCl2 + 50 µL 0.1 M NaOH
○ Gave a pellet

	• 1 mL 4 g/L oxalic acid + 100 µL 5 M KOH + 100 µL 100 mM CaCl2 ○ Immediate precipitation • Same as before but without KOH ○ Light precipitation after about 5-10 minutes • Use oxalate acid directly and pH must be adjusted correctly for proper precipitation pH measurement of the stock solutions • Potassium oxalate 1 mM: pH = 6.6 • CaCl2 100 mM: pH = 5.8

Date	08.09.25
Personal	Daniel Dorta, Amasia Rocha
Protocol	**Overview** This protocol outlines the procedure for the precipitation of calcium oxalate. The experiments described focus on determining the optimal conditions for precipitation, specifically the required pH and molar ratios of reactants. **Materials** • Potassium Oxalate (1 mM stock solution) • Calcium Chloride (100 mM stock solution) • Deionized Water (H2O) • Sodium Hydroxide (0.1 M NaOH) • Oxalic Acid • Potassium Hydroxide (5M KOH) • Sample solutions (unspecified, likely containing oxalate) **Precipitation and Centrifugation** 1. Add potassium oxalate and calcium chloride to a reaction tube. 2. Bring the total reaction volume to 1000 µL with deionized water. 3. Allow the mixture to stand for **30 minutes** to allow precipitation to occur. 4. Centrifuge the samples at **10,000 g for 10 minutes** to pellet the precipitate. **Observations** **Initial Tests** • **Sample 1:** A mixture of 250 µL Sample, 250 µL H2O, and 100 µL of 100 mM CaCl2 resulted in a pellet. This pellet contained both calcium oxalate and other sample components. • **Sample 2:** The supernatant from Sample 1 (250 µL) was mixed with 250 µL H2 O and 100 µL of 100 mM CaCl2. A pellet formed, confirming it was likely pure calcium oxalate.

pH and Reagent Concentration Tests

- **Test 1 (pH Dependence):**
  - Mix 10 µL of 2 mM Oxalic Acid with 890 µL H2O, 100 µL of 100 mM CaCl2, and 50 µL of 0.1 M NaOH.
  - **Observation:** A pellet formed.
- **Test 2 (Alkaline Conditions):**
  - Mix 1 mL of 2 mM Oxalic Acid with 100 µL of 5M KOH and 100 µL of 100 mM CaCl2.
  - **Observation:** Immediate precipitation (flocculation) was observed.
- **Test 3 (Without KOH):**
  - Repeat Test 2 without the addition of KOH.
  - **Observation:** A slight precipitation occurred after 5-10 minutes, indicating that a specific pH is necessary for rapid precipitation.

Concentration Curve Preparation

This section describes a dilution series for a **calibration curve**, likely for a photometric assay to measure remaining calcium or oxalate in solution.

- **Stock solutions:**
  - Potassium Oxalate: 1 mM
  - Calcium Chloride: 100 mM
  - H2O
- **Dilution Series:**
  - Prepare 10 samples with a total volume of 1000 µL each.
  - Vary the volume of Potassium Oxalate and H2O as shown in the table below.
  - The volume of CaCl2 added must provide a specific percentage excess (10%, 50%, or 100%) to ensure all oxalate precipitates. The total volume of CaCl2 added is determined by the desired excess percentage.

Sample #	*1 mM Potassium Oxalate (µL)*	*100 mM CaCl2 (µL)*	*H2 O (µL)*	*Total Volume (µL)*
1	100	110	790	1000
2	80	110	810	1000
3	65	110	825	1000
4	50	110	840	1000
5	35	110	855	1000
6	25	110	865	1000
7	15	110	875	1000
8	10	110	880	1000

9	5	110	885	1000
10	0	110	890	1000

This table is for a 10% calcium excess.

Next Steps

- **New Calibration Curves:**
  - A new calibration curve with oxalic acid will be created.
  - The stock solution will be prepared with the correct pH.
  - The calibration curve will be generated for three different calcium excess percentages:
    - With a 10% excess of calcium.
    - With a 50% excess of calcium.
    - With a 100% excess of calcium.
- **General Information:**
  - The reaction between Calcium and CPC will be observed.
  - Calcium and CPC are needed for the reaction to work correctly.

Date	09.09.2025
Who	Yazan
What	• Stock solution oxalate 1mM 100 mL pH adjusted to pH=7.3 • 3 calibration series prepared with oxalate + H2O and frozen

Date	09.09.25-10.09.25
Personal	Daniel Dorta Bechel, Amasia Rocha
Protocol	**Objective:** To create a calibration curve for a calcium-oxalate precipitation experiment, using a 10% excess of calcium (Ca2+) to ensure complete precipitation of oxalate (C2 O42−). **Materials & Reagents** • 1 mM Oxalate stock solution • 1 mM Calcium Chloride (CaCl2) stock solution • Deionized water (H2O) • Calibration standards (as listed below) **Procedure**

1. Preparation of Oxalate Stock Solution:
    - Adjust a 1 mM Oxalate solution to a pH of 7.5.
2. Preparation of Calibration Standards:
    - Prepare three separate calibration series, each with a total volume of 1000 µL per well.
    - For each series, the amount of oxalate is held constant while the amount of calcium is varied to achieve a 10% calcium excess.
    - The total volume of 1000 µL is made up of a combination of Oxalate, Calcium Chloride, and water.

Calibration Series Setup:

Well #	*1 mM Oxalate (µL)*	*1 mM CaCl2 (µL)*	*H2 O (µL)*	*Total Volume (µL)*
Series 1				
1	100	110	790	1000
2	80	88	832	1000
3	65	71.5	863.5	1000
4	50	55	895	1000
5	35	38.5	926.5	1000
6	25	27.5	947.5	1000
7	15	16.5	968.5	1000
8	10	11	979	1000
9	5	5.5	989.5	1000
10	0	0	1000	1000
Series 2				
1	100	110	790	1000
2	80	88	832	1000
3	65	71.5	863.5	1000
4	50	55	895	1000
5	35	38.5	926.5	1000
6	25	27.5	947.5	1000

7	15	16.5	968.5	1000
8	10	11	979	1000
9	5	5.5	989.5	1000
10	0	0	1000	1000
Series 3				
1	100	110	790	1000
2	80	88	832	1000
3	65	71.5	863.5	1000
4	50	55	895	1000
5	35	38.5	926.5	1000
6	25	27.5	947.5	1000
7	15	16.5	968.5	1000
8	10	11	979	1000
9	5	5.5	989.5	1000
10	0	0	1000	1000

Note: The table reflects the 10% calcium excess for each well.

3. **Well Plate Loading:**
    - Load the prepared standards into a microplate as follows:
        - **Rows A, B, C:** Calibration Series 1 in triplicates.
        - **Rows D, E, F:** Calibration Series 2 in triplicates.
        - **Rows G, H:** Calibration Series 3 in duplicates.

Data Analysis

- Measure the precipitate in each well and plot a calibration curve.
- Note: The lab notes indicate that well #4 of Calibration Series 3 is likely incorrect and should be checked or excluded from the data analysis.

Date	10.09.2025
Who	Amasia, Amelie

What

- 1. Calibration series with 10% Ca excess
  2. Calibration series with 50% Ca excess
  3. Calibration series with 100% Ca excess
- 1000 µL total volume after precipitation:

	1 mM oxalate [µL]	Calibration series 1		Calibration series 2		Calibration series 3	
		H2O [µL]	1 mM CaCl2 [µL]	H2O [µL]	1 mM CaCl2 [µL]	H2O [µL]	1 mM CaCl2 [µL]
1	100	790	110 µL each 1 mM CaCl2	750	150 µL each 1 mM CaCl2	700	200 µL each 1mM CaCl2
2	80	810		770		720	
3	65	825		785		735	
4	50	840		800		750	
5	35	855		815		765	
6	25	865		825		775	
7	15	875		835		785	
8	10	880		840		790	
9	5	885		845		795	
10	0	890		850		800	
	c_{Ca} in Eppi	110 µM		150 µM		200 µL	
	V from the Eppi to the well [µL]	100		73.3		55	
	V_{H2O} in well [µL]	0		26.7		45	
	V_{CPC} into the well [µL]	29.3 µL 7.5 mM oCPC					

Well loading:

	1	2	3	4	5	6	7	8	9	10	11	12
A	Calibration series 1 in triplicates										1	2
B											3	4
C											5	6
D	Calibration series 2 in triplicates										7	8
E											9	10
F												
G	Calibration series 3 in duplicates											
H												

! Sample from calibration curve 3, No. 4 is probably incorrect (pipetting error)

Date	15.09.25
Personal	Daniel Dorta Bechel, Janina
Protocol	Objective:

	Establish reliable Calibration standards from 1-25mM created by optimized RevION_Excel model.Use of 10µl 7,5mM oCPC Stock solution. The Target OD in the TECAN should be about 0,68 because of the dilutions. **Reagents:** • Calcium chloride (CaCl2) • Oxalic acid (Ox) • ddH2O • CPC (Cetylpyridinium chloride) • BCgII (Bromocresol green II) **Step 1: Calibration Curve Preparation** 1. Calibration Series: Create a series of standards with concentrations ranging from 1 to 25 mM. 2. Flocculation: Add CaCl2 to the standards to induce flocculation. 3. Measurement: o Option 1: Measure absorbance using a spectrophotometer (e.g., Tecan) with ddH2O as the blank. o Option 2: Measure absorbance after adding BCgII. **Step 2: Sample Preparation** 1. Sample Series: Prepare samples with concentrations from 1 to 25 mM, without adding CaCl2 for flocculation. 2. Measurement: Measure absorbance of the samples. 3. Dilution: If necessary, dilute samples to ensure measurements are within the linear range of the instrument. *Part 2: Measurement and Data Analysis* **Step 1: Absorption Measurement** 1. Tecan Settings: o Target concentration: 0.68 mM o Dilution: 1:3 2. Procedure: o Measure the absorbance of all standards and samples. o Measure with and without CPC. Note that the measurement "Absorption - Absorption without CPC" might not be functional. **Step 2: Data Analysis** 1. Back-calculation: Use the dilution factor (DF) to back-calculate the "real" absorbance values.

	2. Concentration Calculation: Convert absorbance to concentration (mM) and then to amount of substance (µmol). 3. Mass Calculation: Use the molar mass of oxalic acid (90.03 g/mol) to calculate the mass (m) of the sample. o Example: If n=2×10−6 mol, then m=n×Mm =2×10−6mol×90.03g/mol=1.80×10−4g=0.180mg. **Step 3: Model Verification** 1. Comparison: Compare the results from the two models (Option 1 and Option 2) and assess the fit. 2. Evaluation: Determine if the fit is comparable to the theoretical fit. *Part 3: Centrifugation Test* **Objective:** To optimize the centrifugation process for separating the supernatant. **Procedure:** 1. Initial Attempt: o Centrifuge samples at 5,000 G for 10 minutes. o Observe the supernatant. A failure to clear indicates remaining CaOx. 2. Improved Attempt (Success): o Centrifuge samples at 10,000 G for 15 minutes. o Observe the supernatant. A pellet at the bottom and a clear supernatant indicates a successful precipitation. Note: The orange-labeled samples were successfully separated with this improved centrifugation step. The blue-labeled samples required further dilution. **Kinetics Study:** Kinetic measurements allowed the measurement of real-time complex formation in an incubation time study. This helped in the identification of the maximum oCPC-complexation. - The study was done over a period of 60 minutes. - The result was that measurements should be done as quick as possible; the complex is stable for only about 10-20 minutes.

Date	22.09.25
Personal	Daniel Dorta Bechel
Protocol	**Objective:** To test the effectiveness of calibration standards in a CGXII media solution.

	Materials: • CGXII media • Oxalate stock solution • Calcium calibration standards • Deionized water • Test tubes • Spectrophotometer **Procedure:** 1. Preparation of Standards: ○ Prepare a series of calcium calibration standards by diluting the stock solution with deionized water. ○ Label each standard with its corresponding calcium concentration. 2. Blank Preparation: ○ Prepare a blank sample containing only CGXII media. This will serve as a negative control. 3. Sample Preparation: ○ Add a measured amount of oxalate stock solution to the calibration standards. This will induce the expected calcium-oxalate precipitation. ○ Add the same amount of oxalate to the blank sample. 4. Reaction and Measurement: ○ Mix all samples thoroughly. ○ Incubate the samples for a specified time to allow for complete precipitation. ○ Measure the turbidity or absorbance of each sample using a spectrophotometer to quantify the precipitation. **Discovery** During the experiment, an unexpected precipitation was observed in the blank sample containing only CGXII media. Further research revealed that the added calcium present in the CGXII media does not only precipitate with the added oxalate but also with the phosphate ions naturally present in the media. This previously unaccounted-for reaction significantly impacted the results, as it introduced an additional source of precipitation independent of the oxalate addition.

Date	24.09.25
Personal	Daniel Dorta Bechel, Lea Wührl
Protocol	Objective:

	Use of Sigma-Aldriche´s Enzymatic Oxalate Assay Kit, to analyze our samples. Protocol was followed as instructed by the provider. Sample bank (DasGip experiment 23.07.2025) [max.8 Samples per reactor for analysis] - Reactor1 (pPBEx2_oahA) (IPTG) [t=1,t=3,…,t=15] - Reactor2 (pPBEx2_oahA) (IPTG) [t=1,t=3,…,t=15] - Reactor3 (Wild Type) [t=0,t=2,…,t=10] - Reactor5 (pPBEx2_oahA) [t=1,t=3,…,t=13] 1. Sample Transfer • Step 1: Transfer 100µL of your sample into a new Eppendorf tube. • Step 2: Before proceeding, thoroughly mix the tube using a vortex. 2. Centrifugation • Step 3: Centrifuge the sample at 10,000×g for 10 minutes. 3. Well Preparation • Step 4: Pipette 10µL of sample into the designated well(s). • Step 5: Pipette 10µL of HPLC water into the blank and sample wells. • Step 6: Pipette 10µL of the standard into the internal standard wells. 4. Adding Reagents • Step 7: Add 130µL of the standard reagent. • Step 8: Add 30µL of blank reagent. • Step 9: Add 30µL of other reagents. 5. Plate Loading and Measurement • Step 10: Load the plate. Results: The pPBEx2_oahA IPTG induced culture from reactor one showed semi-promising results, the result curve had some variance between the samples. Our WT from reactor three also showed some semi-promising results, in comparison to our culture in R1 we can deduce that our engineered strains are producing more oxalate than before.

3 Primer and plasmids

Primer

Name	Purpose	Sequence
SB1_pGiga_oahA_for	oahA-fragment construction for subcloning in pEKEx2	5‘-cat gat tac gcc aag ctt gc-‘3
SB2_pGiga_oahA_rev	vs	5‘-cgg gga tcc tct aga gtc g-‘3
SB3_GA_oahA_for	oahA-fragment construction for pEKEx2_oahA assembly	5‘-gcc aag ctt gca tgc ctg cat ttg ttt aac tt-‘3
SB4_GA_oahA_rev	vs	5‘-tac ccg ggg atc ctc tag agt caa tgg tga tga tgg tgg-‘3
SB5_CG_seq_oahA_for	sequencing of oahA-fragments in pEKEx2_oahA	5‘-gat atg acc atg att acg cc-‘3
SB6_CG_seq_oahA_rev	vs	5‘-ctt ggc tgt aaa acg acg gc-‘3
SB7_Fragment 1_For	Upstream-homologue construction for pK19_oahA assembly	CCTGCAGGTCGACTCTAGAG tccgtgaaatgtggccaggc
SB8_Fragment 1_Rev	vs	agctcatttcaaacggatgggcacct tttaatagc
SB9_Fragment 2_For	oahA-fragment construction with Ptac und lac-operon for pK19_oahA assembly	ccatccgtttgaaatgagctgttgac aattaatcatcgg

(SB10_Fragment 2_Rev)	vs *flawed, since it was mitakenly constructed for unoptimised oahA*	gtgcccatccTCAATGGTGATG ATGGTGAGA
(SB11_Fragment 3_For)	Downstream-homologue construction for pK19_oahA assembly *flawed, since it was mitakenly constructed for unoptimised oahA*	TCCGCCATTGAggatgggcacct tctcgttt
SB12_Fragment 3_Rev	vs	TTGTAAAACGACGGCCAGTG gtttgttatatccgccgagtagcaag
Fragment 2_Rev_new	See above; new, since Fragment 2_Rev was flawed	gtgcccatccTCAATGGTGATG ATGGTGGTGT
Fragment 3_For_new	See above; new, since Fragment 3_For was flawed	TCACCATTGAggatgggcacctt ctcgttt
SB13_Forward inner seq	Primer for sequencing of Ptac-lac-oahA-fragment in *C. glutamicum* gDNA after genome integration	ctcgttctgcagcgg
SB14_Reverse Inner Seq	vs	gtcccaaattgaaagtgctg
SB15_Forward Outer Seq	vs	cttcctgaccaccccaacc
SB16_Reverse outer seq	vs	cgggctcgcttaatttcag
SB17_Fragment 1 for	Upstream-homologue construction for pK19_oahA assembly	CTATGACCATGATTACGCCA ctcaaggtcatccgtgaaatgtgg
SB18_Fragment 1 rev	vs	ccatgccggccaaacggatgggcac c
SB19_Fragment 2 (lacI) for	lacI-fragment construction for pK19_lacI_oahA assembly	catccgtttggccggcatggcggc
SB20_Fragment 2 (lacI) rev	vs	gcgcaaaaaacattatccagaacgg g
SB21_Fragment 3 (oahA) for	oahA-fragment construction with Ptac und lac-operon for pK19_lacI_oahA assembly	ctggataatgttttttgcgccgac
SB22_Fragment 3 (oahA) rev	vs	gtgcccatccTCAATGGTGATG ATGGTGGTGTTC

SB23_Fragment 4 for	Downstream-homologue construction for pK19_lacI_oahA assembly	TCACCATTGAggatgggcacctt ctcg
SB24_Fragment 4 rev	vs	TCCTCTAGAGTCGACCTGCA gcttctttgaagagtctcgctgtt
SecP1	pK19 control	TAATGCAGCTGGCACGAC
SecP2	vs	CCATGTAAGCCCACTGCA

Plasmids

Plasmid	Purpose	Source
pGiga_oahA	Vector containing oahA-gene for subcloning into an expression vector	Gigabases
pEKEx2	Expression vector	Bakkes et al.
pPBEx2	Improved pEKEx2-derived expression vector	Bakkes et al.
pEKEx2_oahA	oahA-Expressioin-Vector	This work
pPBEx2_oahA	oahA-Expression-Vector	This work
pK19mobsacB	Genome integration vector	Institute for Biochemical Enigneering, Stuttgart

Literature:

Bakkes et al. (2020): Improved pEKEx2-derived expression vectors for tightly controlled production of recombinant proteins in Corynebacterium glutamicum. Plasmid, Volume 112, 102540. *https://doi.org/10.1016/j.plasmid.2020.102540*